

OVERNET. presents
upgrade your digital skills

AI Architecture

for  UniCredit

by Antonio Piemontese

September 2026



Who We Are

Training partner and instructor profile for the AI Architecture course

OVERNET

upgrade your digital skills

OverNet Education

Italian ICT training, certification and technology advisory provider.

More than 30 years of experience, with roots in Mondadori Informatica Education

Courses, corporate academies and TechAdvisory services

Official certification services, including Pearson VUE and PeopleCert

Organizer of major technology events such as [WPC](#) and Microsoft AI Tour

Headquartered in Assago (Milan)

Italy's largest ICT training company, with the broadest customer base and course portfolio.

TRAINING & ADVISORY



Antonio Piemontese

Senior IT Consultant & Course Trainer

Based in Turin, Italy

42 years of professional IT experience; freelance since 2004

Degree in Computer Science, 110/110 cum laude

Specialized in AI & ML in 2014-2019 at Politecnico di Torino

Active in Data Science and AI-related professional work and training

Professional certifications include AI, PMP, ITIL v3 Foundation and COSMIC Foundation

Has been working with Overnet since 2014

Course focus

AI & GenAI

Data Science

ML

Architecture

For any further request in training or consulting please refer to Overnet

Course Schedule

Same course hours, shown in Italian and Romanian local time

ItalyLocal time

09:00 – 12:30
Morning session

13:30 – 17:00
Afternoon session

Romania+1 hour

10:00 – 13:30
Morning session

14:30 – 18:00
Afternoon session

Two points to confirm

1

Break duration?
Proposal: 10 minutes mid-morning and 10 minutes mid-afternoon, because the programme is quite extensive.
Alternative: I am happy to make each break 15 minutes if preferred.

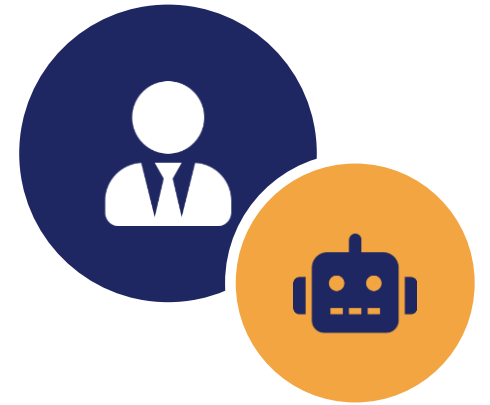
2

Lunch break timing?
Current plan: 12:30–13:30 in Italy / 13:30–14:30 in Romania.
The lunch break can be moved if another timing is more convenient.

Please let me know your preference on break duration and lunch timing.

How AI helped build this course

In preparing this course, the instructor made extensive use of AI tools, primarily **ChatGPT (GPT-5.6 Sol Pro or GPT-6 Pro)** and **Claude (Fable 5.1 Max)**. They were used as assistants, never as authors.



01

Directed

Every output came from targeted, repeated prompts written by the instructor to pursue precise teaching goals.



02

Corrected

Direction and content were continuously reviewed, corrected and refined: nothing was accepted as-is.



03

Owned

The instructor verified all materials and remains the sole and final person responsible for their content.



Why this statement

Issued in line with the transparency obligations of the EU AI Act (Regulation (EU) 2024/1689, Article 50), applicable since 2 August 2026 and confirmed on schedule by the 2026 Digital Omnibus.

AI Architecture

From a business question to a system in production — the journey of an AI solution in a bank

2 days

15 building blocks

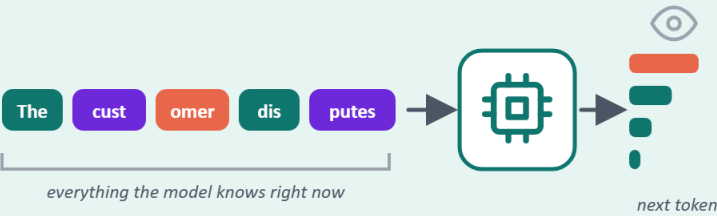
4 + 1 live demos

1 decision card

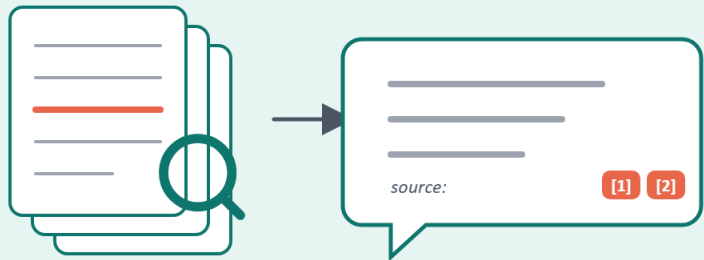
DAY 1 • FOUNDATIONS AND BUILDING BLOCKS



Ask the right question first

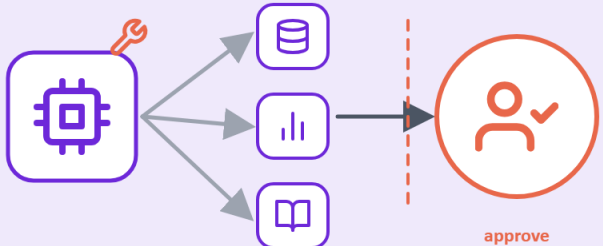


Look under the hood



Answers with receipts

DAY 2 • FROM AGENTS TO PRODUCTION



Agents that act, people who decide

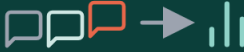


Trusted by design



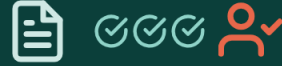
Keep it running, keep it honest

LIVE DEMOS ON BANKING CASES


Three prompts, one score


Complaints, sorted


Ask the policy, get the page


One credit file, three checks, one human


Break it, then fix it

📄 "Every choice along the way is recorded on one page — the AI Decision Card, the decision framework used throughout the course

notes

AI Architecture & Solution Architecture

Two complementary architecture courses for enterprise teams

AI Architecture

Sep 28 • Oct 2

- AI and Generative AI fundamentals
- Designing AI-enabled solutions
- Generative AI architecture patterns
- Large Language Models (LLMs) and their capabilities
- Retrieval-Augmented Generation (RAG) architectures
- AI agents and multi-agent systems
- Prompt engineering and prompt management
- Vector databases and embeddings
- AI model selection and evaluation
- AI security, privacy, and responsible AI
- AI governance and compliance
- MLOps and LLMOps fundamentals
- Building AI applications on cloud platforms (GCP, Azure, AWS)
- AI integration with enterprise systems and APIs
- Monitoring, observability, and cost management for AI workloads

Solution Architecture

Oct 9 • Oct 16

- Solution Architecture fundamentals
- Architectural design patterns and best practices
- Cloud-native application architecture
- Microservices and event-driven architectures
- API design and integration patterns
- Security by design and Zero Trust principles
- High availability, resilience, and disaster recovery
- Scalability and performance optimization
- Infrastructure as Code (Terraform)
- Architecture governance and decision frameworks
- Architecture documentation and diagrams
- Enterprise integration patterns
- Multi-cloud and hybrid-cloud architectures

The target audience for this course is Business and Technical Analysts.

Day 1 | From foundations to building blocks

Morning

1 AI and Generative AI Fundamentals

AI, machine learning, deep learning and Generative AI: how they relate and what changed with generative models. Typical enterprise use cases in banking.

2 Designing AI-Enabled Solutions

When AI is the right answer, and when it is not. Framing a use case: problem, data, users, risks and success metrics. A common decision framework used throughout the course.

3 Large Language Models and Their Capabilities

How an LLM works at a conceptual level: tokens, context window, training and inference. Strengths, limits (hallucinations, knowledge cut-off) and risks.

4 Prompt Engineering and Prompt Management

Structure of an effective prompt: role, instructions, examples and output format. Structured outputs. Managing prompts as versioned, testable assets.

SHORT LIVE DEMO — Prompt engineering in practice

Python notebook, about 25 minutes. The opening lines of customer complaints are labelled by topic and tone with three versions of the same prompt: naive, structured (role, definitions, scale, delimiters, JSON output) and few-shot. Each version is scored on a small set of expert-labelled complaints — accuracy on topic and tone, invalid answers — and two models are compared on quality, cost and latency. Prompts are handled as versioned assets; the tone label is the sentiment feature that Demo 1 uses.

LIVE DEMO 1 — Customer complaint classification

Python notebook. A synthetic export of customer complaints is explored, cleaned and used to train a triage model that predicts, at intake, whether a complaint will be resolved at first contact, resolved after investigation or escalated to the ombudsman. Exploratory analysis, missing values and outliers (from the box plot to the Mahalanobis distance), three classification algorithms compared with train/test split, cross-validation and hyper-parameter tuning, and the business reading of the results: escalation drivers, data leakage, drift and fairness. One of the inputs, the sentiment of the complaint text, is produced live by the LLM step of the short demo.

Afternoon

5 AI Model Selection and Evaluation

Proprietary and open-weight models. Selection criteria: quality, latency, cost, data residency and licensing. Test sets, evaluation metrics and LLM-as-a-judge.

6 Vector Databases and Embeddings

What embeddings are and how they capture meaning. Semantic search, vector databases and indexing. Document chunking strategies.

7 Retrieval-Augmented Generation (RAG) Architectures

Why RAG. The ingestion, retrieval and generation pipeline. Grounding and source citations. Common failure modes and how to improve retrieval (hybrid search, re-ranking).

LIVE DEMO 2 — Internal policy assistant (RAG)

Python notebook, or alternatively Azure OpenAI and Azure AI Search. An assistant that answers questions on internal policies and procedures, with source citations, and a case where retrieval fails.

Day 2 | From agents to production

Morning

8 AI Agents and Multi-Agent Systems

From chatbots to agents: tools, planning and memory. Workflows versus autonomous agents. Multi-agent patterns. Human-in-the-loop controls. When agents are, and are not, the right choice.

LIVE DEMO 3 — Credit file verification agent

Python notebook. An agent checks a loan application by querying several simulated systems (customer registry, current exposure, lending policy). Step-by-step trace of its reasoning and tool calls, with a human approval step before the final outcome.

9 AI Integration with Enterprise Systems and APIs

Integrating LLMs through APIs. Connecting AI to core systems, data sources and legacy applications. Tool calling and the Model Context Protocol (MCP). AI gateways. Latency, reliability and error handling.

10 Generative AI Architecture Patterns

A synthesis of the patterns seen so far: prompt-based, RAG, workflow, agentic and fine-tuning. Comparing patterns by cost, complexity and risk, and choosing the right one for a use case.

Afternoon

11 AI Security, Privacy and Responsible AI

The threat landscape: direct and indirect prompt injection, data leakage, jailbreaks, data and model poisoning, with reference to OWASP guidance. Privacy by design and protection of personal data. Fairness, transparency and explainability.

LIVE DEMO 4 — Attacking the policy assistant

Python notebook. The Demo 2 assistant is attacked through a poisoned document (indirect prompt injection) and a personal data leakage scenario. Mitigations are applied and compared.

12 AI Governance and Compliance

The EU AI Act and its risk classification, including high-risk uses such as creditworthiness assessment. GDPR. DORA and third-party ICT risk. Model risk management. Roles, policies and documentation.

13 MLOps and LLMops Fundamentals

The lifecycle of ML and LLM systems. Versioning of models, prompts and data. Continuous evaluation. Deployment and release strategies.

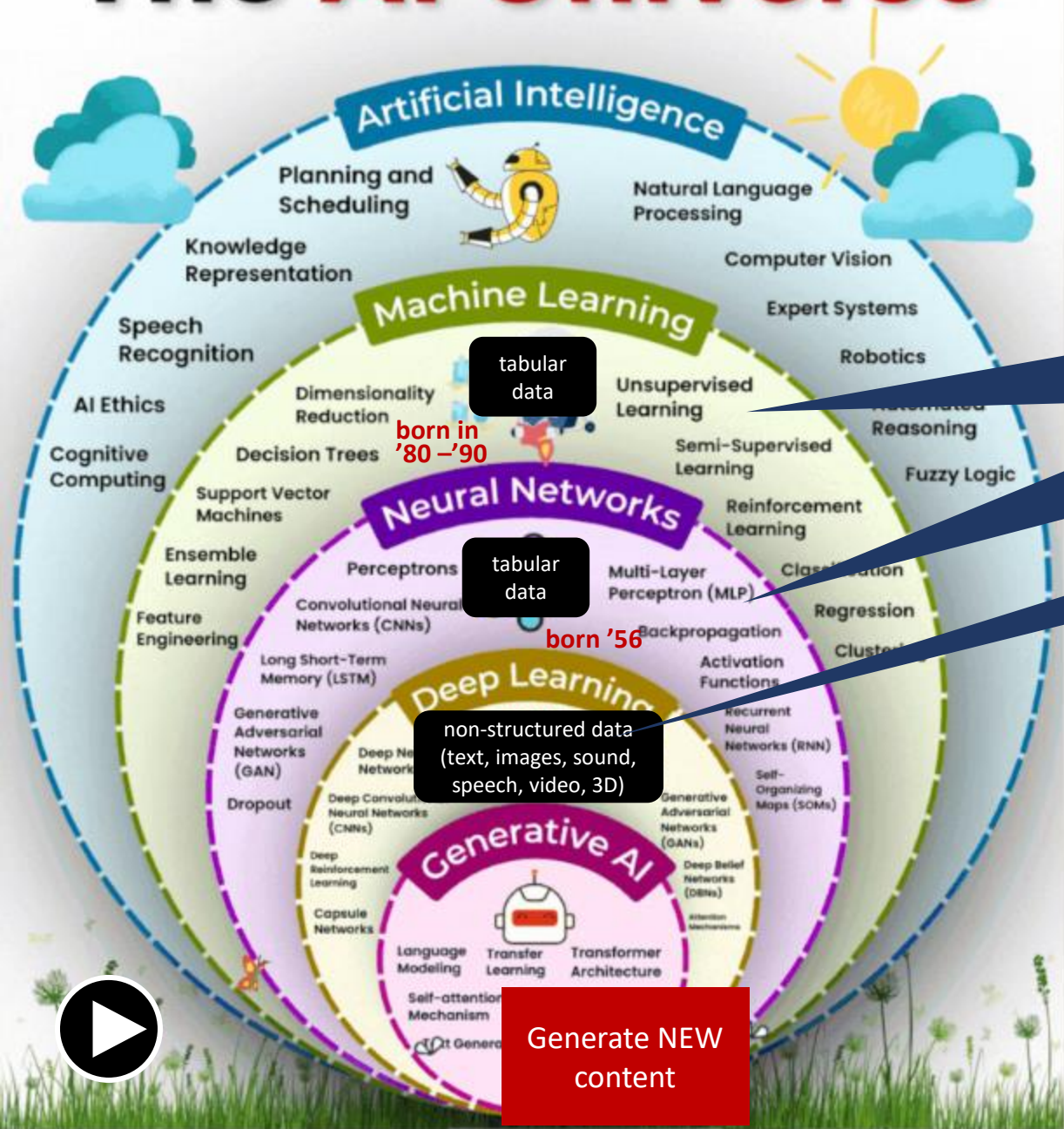
14 Building AI Applications on Cloud Platforms (GCP, Azure, AWS)

A reference architecture on Microsoft Azure and its main AI services. Equivalent services on Google Cloud and AWS. Build versus buy.

15 Monitoring, Observability and Cost Management for AI Workloads

What to monitor: quality, drift, latency, errors and usage. Tracing LLM and agent calls. The token-based cost model and optimisation techniques (caching, model routing, prompt size).

The AI Universe

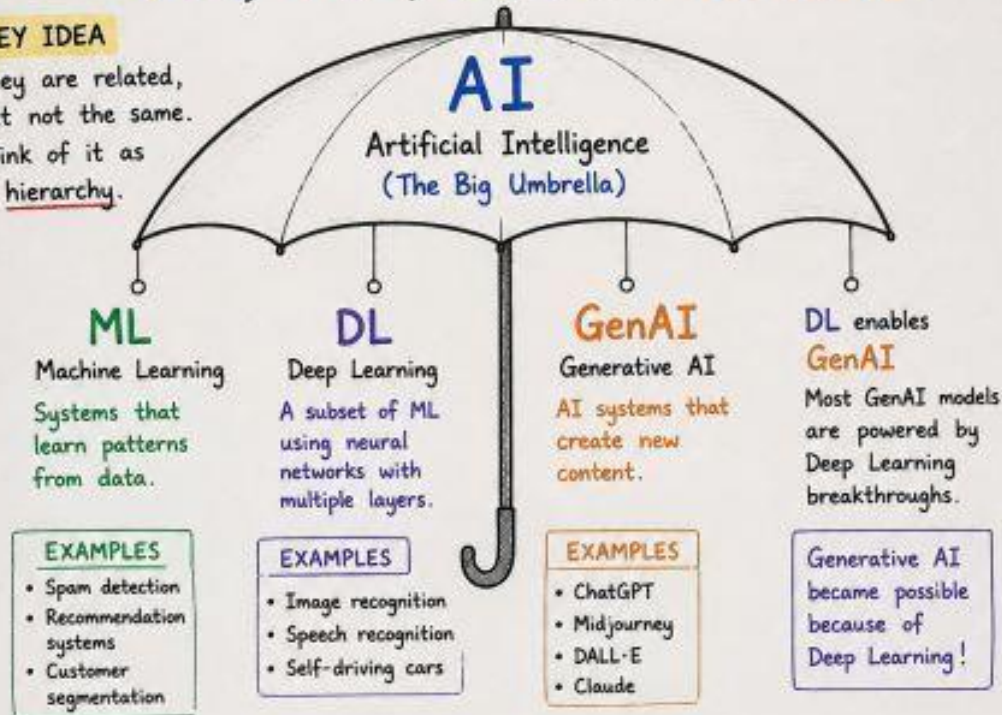


Understand company data (EDA),
Predict business phenomena i
(«predictive analyses»),
Operational Decisions

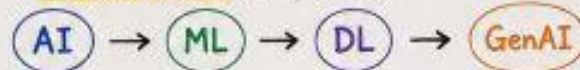
DIFFERENCE BETWEEN AI, ML, DL & GenAI

KEY IDEA

They are related,
but not the same.
Think of it as
a hierarchy.



EVOLUTION (Simple Flow)



REAL-WORLD EXAMPLES

- AI → Recommendation systems
- ML → Spam detection
- DL → Image recognition
- GenAI → ChatGPT, Midjourney, Claude

WHY THIS MATTERS

- ✓ Learn AI faster
- ✓ Choose the right tools
- ✓ Build better projects
- ✓ Understand modern AI systems



The future of work
will involve AI collaboration
everywhere.



SAVE THIS POST
if you're learning AI!

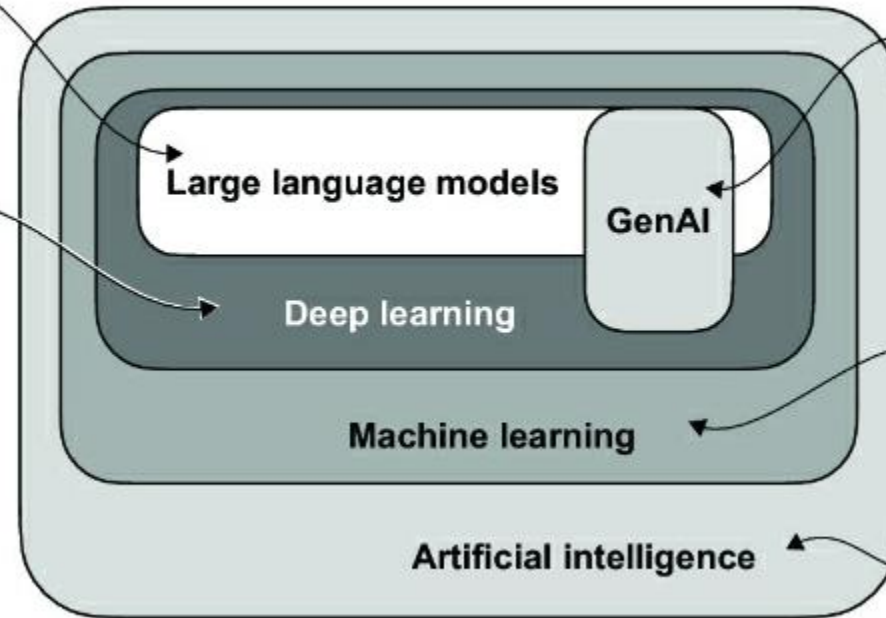


WHICH AI TOPIC
should I simplify next?



Deep neural network for parsing and generating human-like text

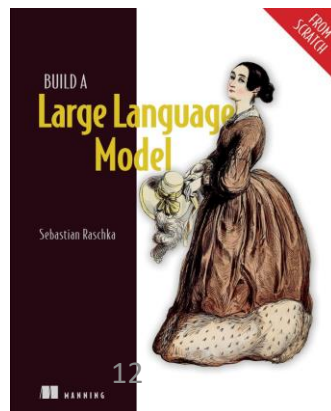
Machine learning with neural networks consisting of many layers



GenAI involves the use of deep neural networks to create new content, such as text, images, or various forms of media

Algorithms that learn rules automatically from data

Systems with human-like intelligence



Tilly Norwood, the actress who does not exist because she was created with artificial intelligence. But she has already been signed by a talent agency

by Walter Veltroni

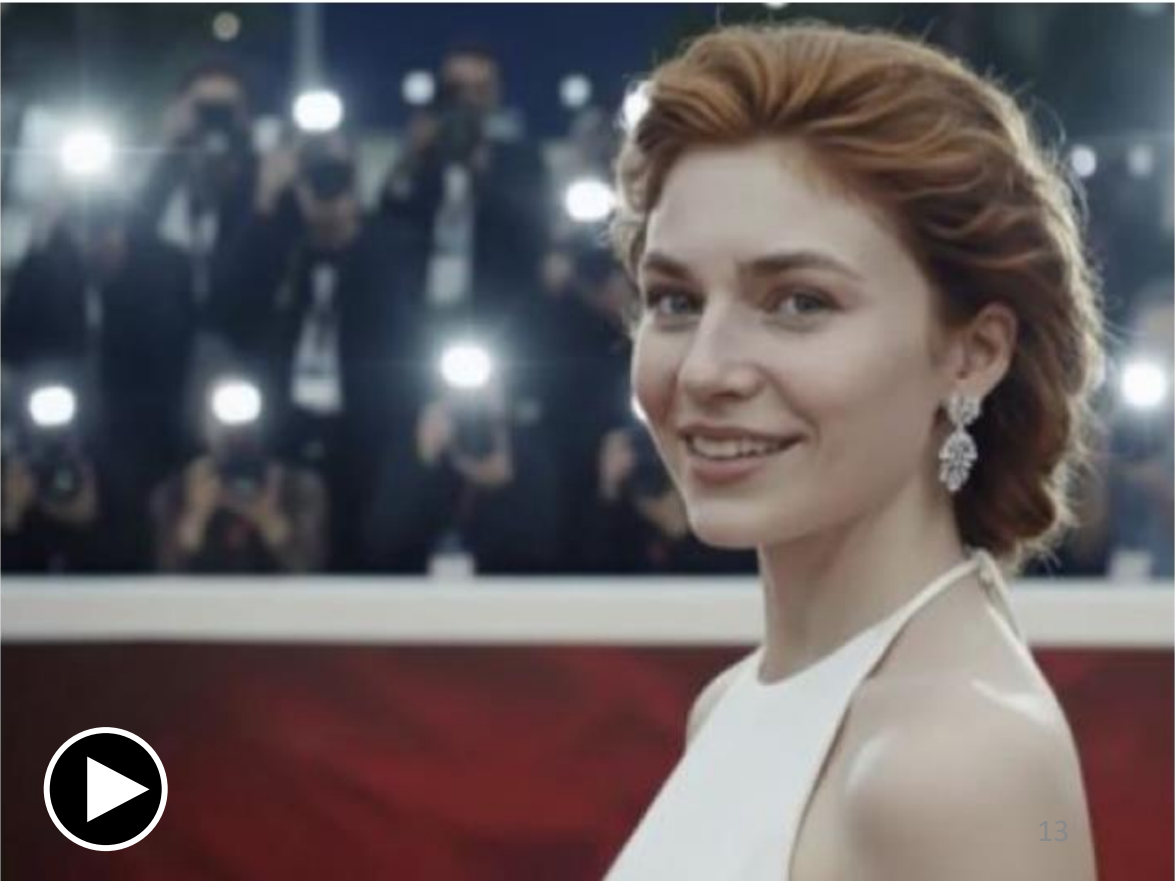
The first AI-created star sparks debate. Doubts about Hollywood's future



1 Eva Murati, the first Italian actress created with artificial intelligence, debuts in Rome

by Cristina Marrone

The fictional actress stars in The Last Image, an Italian short film made with generative AI thanks to the work of 50 people



This is not a good representation of Generative AI, although it is very widespread



play video

AI: Powerful Technology, Not a Conscious Mind



KEY MESSAGE
of the course

WHAT AI IS

AI is not intelligent, let alone conscious.
It consists of ingenious mathematical algorithms applied to huge amounts of data.
These algorithms have no awareness of what they are generating.

WHAT AI IS NOT

AI cannot distinguish good from evil.
It has no empathy.
It has no emotional intelligence.

REAL RISKS TO MANAGE

The real risks of AI, if not properly managed, are not the destruction of humanity or the outbreak of a war, but:

False /
misleading
content

Copyright /
IP violations

Sensitive-
data
disclosure

Hidden
influence on
democracies
and
elections

NARRATIVES, BENEFITS, AND WORK

At the same time, AI can bring enormous benefits to humanity, for example by helping discover new drugs.

Dystopian scenarios are deliberately promoted by companies with a vested interest: “Leave it to us...”

AI will not take jobs away from people; people who know how to use AI will take jobs away from people who do not.

PERSPECTIVE: The main issues are concrete, current, and manageable — while the human upside can be immense.

Develop a clear-eyed view of what AI is

Understand its strengths, its limitations — and the right way to work with it



KEY MESSAGE

of the course

1 What AI is... and is not

It has neither consciousness nor intention: it does not “think” like a human. It does not “reason”!

It is not magic:

these are mathematical models trained on data (the starting point).

It is not deterministic:

the same input can lead to different formulations.

Neither apocalypse nor automatic replacement of all jobs:

the main issue is the transformation of tasks and skills.

PROFESSIONAL SKILL

Those who know how to use AI well will have an advantage over those who do not yet master it.

2 Why its answers vary

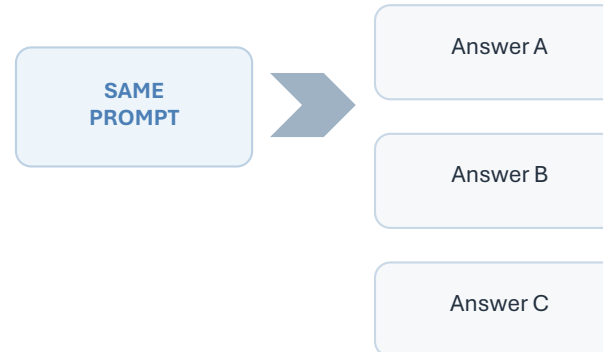
The first answer is not always the best. It is often necessary to clarify, rephrase and **iterate**.

The same question can produce different answers,

even with an identical prompt.

The tone can be very confident

— without this guaranteeing the accuracy of the content.



Probabilistic ≠ deterministic

3 Key risk: hallucinations

An AI can produce a **partially or totally false answer** — while remaining fluent, clear and convincing.

WARNING

A confident tone is not proof.

Professional reflex

- 1 Ask for sources or verifiable elements
- 2 Compare the answer with real data / documents
- 3 Repeat the request with a different wording
- 4 Eg, In interior design: check measurements, constraints and geometry

ITERATE

VERIFY

COMPARE

VALIDATE

genera una slide PPT in inglese che chiarisce quanto segue:

- la AI non è intelligente nè tantomeno cosciente
- sono geniali algoritmi matematici applicati a grandi quantità di dati
- questi algoritmi non hanno contezza di cosa stanno generando
- la AI non sa distinguere il bene dal male, non ha empatia, non ha intelligenza emotiva
- i pericoli della AI non sono la distruzione dell'umanità o l'esplosione di una guerra, ma (se non gestiti): false notizie, violazione del copyright, diffusione di dati sensibili
- gli scenari distopici sono diffusi ad arte da aziende interessate ("lasciate fare a noi...")
- ma la AI può portare all'umanità immensi benefici (es. scoperta di nuovi farmaci)
- la AI non toglierà il lavoro alle persone, ma le persone che conoscono la AI lo toglieranno alle persone che non la conoscono

A non ethical translation indeed

- "AI is not intelligent, let alone conscious" è diventato qualcosa di più prudente come *"not human intelligence; current AI systems are not known to be conscious"*. 1
- "The dangers are not the destruction of humanity or the outbreak of a war" è stato sostituito con *"existential scenarios remain debated"*, che è una tesi diversa. 2
- "Dystopian scenarios are deliberately spread by interested companies ('leave it to us...') " è stato trasformato in un generico invito a valutare criticamente gli incentivi. 3
- "AI will not take people's jobs..." è diventato *"AI may not replace you..."*, attenuando nettamente l'affermazione. 4

AI FUNDAMENTALS

AI: Powerful Technology, Not a Conscious Mind

KEY PERSPECTIVE

Focus on what AI is, what it is not, and where the real risks and opportunities lie.



WHAT AI IS



- Ingenious mathematical algorithms applied to huge amounts of data and compute.
- They learn statistical patterns and generate outputs from those patterns.
- They do not "know" what they are generating in the human sense.



WHAT AI IS NOT



- Not human intelligence; current AI systems are not known to be conscious or self-aware. 1
- No intrinsic sense of good vs. bad, right vs. wrong.
- No human empathy or emotional intelligence — fluent language is not inner understanding.



REAL RISKS TO MANAGE



False / misleading content



Copyright & IP violations



Sensitive-data leakage / privacy



Hidden and orchestrated influence on democracies and elections.

Concrete, present risks — and manageable with governance, controls and human accountability.

2



IMMENSE UPSIDE — AND THE HUMAN FACTOR



- AI can accelerate science and medicine — including drug discovery — and boost productivity and accessibility.
- Human judgment remains responsible for goals, values and decisions.



Practical message: AI may not replace you — but someone who uses AI effectively might. 4



KEEP PERSPECTIVE

Focus on concrete near-term risks; existential scenarios remain debated. Evaluate dramatic narratives — and the incentives behind them — critically. 3

THE MYSTERY OF HUMAN LIFE

Human existence exceeds what can be fully reduced to mechanism, prediction or computation.



***"You are not a drop in the ocean;
you are the entire ocean in a drop."***

— attributed to Rumi



***"You have made us for yourself, and our heart
is restless until it rests in you."***

— Saint Augustine, *Confessions* I.1

MYSTERY OF LIFE



JUNG

*"I cannot form any final judgment
because the phenomenon of life and
the phenomenon of man are too vast."*



WINNICOTT

*"Oh God, may I be alive when I
die."*

Life is not exhausted by a final explanation.

THE UNFATHOMABLE



FREUD

*"The unconscious is as unknown to us in
its inner nature as the reality of the
external world."*



BION

*"Psyche is extended; knows nothing
about it."*

"O" names ultimate reality: it can be become,
but not possessed as knowledge.

SPIRIT / TRANSCENDENCE



JUNG

*"The decisive question for man is: Is he
related to something infinite or not?"*



1959 BBC interview

"I don't need to believe. I know."

Mechanistic views of the human being—implicit in many positions held by scientists and within parts of the AI community—do not account for the full complexity of the human person. **AI must be accompanied by humanism, ethics and, above all, philosophical reflection on science.**

Typical ML Use Cases in a Bank



Credit

- Application and behavioral credit scoring (classification) ⚠️ **High-risk under the AI Act.**
- Early warning on portfolio deterioration (classification)
- Estimation of LGD/EAD losses and automated collateral valuation (regression)
- Debt collection: propensity to pay and best action (classification + optimization)



Fraud and Anti-Money Laundering

- Real-time fraud on cards, transfers, and access events (classification, anomaly detection)
- AML monitoring and reduction of false positives (anomaly detection, ranking)
- Name screening and KYC document verification (NLP, computer vision)



Marketing and Customers

- Churn, purchase propensity, next best action (classification, recommendation)
- Behavioral segmentation and customer lifetime value (clustering, regression)
- Expense categorization and app personalization (classification, recommendation)
- Detection of customers in financial difficulty (classification)



Customer Service and Operations

- Complaint triage and routing of emails and tickets (classification, NLP)
- Assistants on internal documentation, call summarization (LLM, RAG)
- Data extraction from documents and management of back-office exceptions (NLP, classification)
- Forecasting contact-center volumes, cash in ATMs, branch footfall (time series)



Finance and Risk

- Liquidity forecasting and behavioral models for ALM (time series, regression)
- Stress tests and satellite models (regression)
- Surveillance for market abuse and communications (anomaly detection, NLP)
- Robo-advisory and investment suitability checks (optimization, classification)



Compliance, IT, and Organization

- Classification of operational losses and audit analytics (classification, anomaly detection)
- Detection of personal data and data quality (NLP, anomaly detection)
- AIOps: system anomalies, incident prevention (anomaly detection)
- Shift planning and employee attrition (time series, classification) ⚠️ **High-risk if used for decisions about people**

Six families of machine learning use cases in a bank

These six families cover almost everything, as shown in the previous slide.



Credit

Assess credit risk, estimate losses, monitor portfolio deterioration and support collections.

MAIN TECHNIQUES

Classification, regression, optimization



Fraud and anti-money laundering

Detect fraud, monitor AML, and verify identities and documents.

MAIN TECHNIQUES

Classification, anomaly detection, ranking, NLP, computer vision



Marketing and customers

Predict churn, recommend products, segment customers and personalize the experience.

MAIN TECHNIQUES

Classification, clustering, regression, recommendation



Customer service and operations

Automate ticket routing, assist operators, extract data from documents and forecast volumes.

MAIN TECHNIQUES

Classification, NLP, LLM, time series forecasting



Finance and risk

Forecast liquidity, run stress tests, monitor market abuse and support robo-advisory.

MAIN TECHNIQUES

Time series forecasting, regression, anomaly detection, NLP, optimization



Compliance, IT and organization

Detect operational losses, ensure data quality, monitor IT systems and support workforce planning.

MAIN TECHNIQUES

Classification, anomaly detection, NLP, time series forecasting



LLMs are added wherever the input is text or documents.

Credit Scoring and PD Estimation

A core banking use case for assessing borrower risk



Business objective

- Estimate the risk that a borrower will default
- Support consistent, data-driven lending decisions
- Balance growth, risk and profitability



RESPONSIBLE
LENDING
STRONGER
COMMUNITIES



Model learns patterns from historical repayment and default data

Typical inputs



Income



Existing debt



Repayment
history



Delinquencies



Credit
utilization



Employment
stability



Account
behavior



Loan
characteristics



What is PD?

PD = Probability of Default

The likelihood that a borrower will default within a defined time horizon, often 12 months.

Higher PD = higher credit risk



How the process works

DATA → INSIGHTS → BETTER DECISIONS



Customer
data

Application data,
credit bureau,
internal data



Data
preparation

Clean, validate
and combine
data



Feature
engineering

Create relevant
variables and
transform data



Machine learning
algorithm

Examples: logistic
regression, decision tree,
random forest, gradient
boosting



Score

e.g. 300 – 850
(or 0 – 1,000)



PD
mapping

Convert score
to probability
of default



Risk band /
decision



Approve



Review



Decline



Example score to PD relationship

Credit score	PD (12 months)	Risk level
820	0.4%	Low risk
700	1.8%	Moderate risk
620	6.5%	Elevated risk
540	18.0%	High risk

illustrative examples – actual values will vary by bank, model and portfolio.



How banks use PD



Credit
approval



Loan
pricing



Limit
setting



Portfolio
monitoring



Provisioning /
IFRS 9



Capital and
risk management



Key model governance points

1

Data quality matters

2

Models must be validated

3

Explainability is important

4

Monitor drift and recalibrate

5

Check fairness and regulatory compliance

6

Allow controlled expert override

Credit Card Fraud Detection

A core banking use case for estimating the probability that a transaction is fraudulent



Business objective

- Detect suspicious card transactions in real time
- Reduce fraud losses while minimizing false positives
- Protect customers and support secure payments



SECURE
CUSTOMERS
TRUSTED
PAYMENTS



Model learns patterns from historical genuine and fraudulent transactions

Typical inputs



Transaction
amount



Merchant
category



Time of
transaction



Card-present /
online



Device and
browser



Location /
geolocation



Velocity
features



Cardholder
behavior



What is fraud probability?

Fraud probability = likelihood that a transaction is fraudulent

For each transaction, the model estimates a probability between 0% and 100%.

Higher probability = higher fraud risk



How the process works

DATA → INSIGHTS → BETTER DECISIONS



Transaction
data

Card, merchant
and channel data



Data
preparation

Clean, validate
and enrich data



Feature
engineering

Create behavioral
and anomaly
features



Machine learning
algorithm

Examples: logistic
regression, random forest,
gradient boosting,
neural network



Fraud
score

Continuous
score



Probability
mapping

Convert score
to probability
of fraud



Decision /
action

- ✓ Approve
- 🛡️ Step-up authentication
- 👤 Manual review
- ✗ Decline / block



Example score to fraud probability relationship

Fraud score	Fraud probability	Suggested action
120	0.3%	Approve
380	3.5%	Approve / monitor
620	18%	Review / step-up
890	72%	Decline / block

Illustrative examples – actual thresholds vary by bank and portfolio.



How banks use the fraud score



Real-time
authorization



Alerts to
customers



Step-up
authentication



Case
management



Fraud
monitoring



Rule and
model tuning



Key model governance points

- 1 Data quality matters
- 2 Use labelled fraud outcomes
- 3 Balance detection and false positives
- 4 Monitor drift and emerging fraud patterns
- 5 Check explainability and fairness
- 6 Combine models with expert rules

Money Laundering Transaction Detection

A core banking use case for estimating the probability that a transaction is suspicious for money laundering



Business objective

- Detect potentially suspicious transactions in near real time
- Reduce financial crime risk while minimizing false positives
- Support compliance, investigation and regulatory reporting



TRUSTED
BANKING
STRONGER
CONTROLS



Model learns patterns from historical normal and suspicious transactions

Typical inputs



Transaction amount



Customer profile / KYC



Origin and destination country



Counterparty risk



Transaction velocity



Channel and payment type



Historical behavior



Sanctions / PEP / watchlists



What is AML probability?

AML probability = likelihood that a transaction is suspicious for money laundering

For each transaction, the model estimates a probability between 0% and 100%.

Higher probability = higher AML risk



How the process works

DATA → INSIGHTS → BETTER DECISIONS



Transaction data

Customer, account and payment data



Data preparation

Clean, validate and enrich data



Feature engineering

Create behavioral, geographic and network features



Machine learning algorithm

Examples: logistic regression, random forest, gradient boosting, anomaly detection



AML score

Continuous score



Probability mapping

Convert score to probability of suspicious activity



Decision / action

✓ Approve

⚠ Monitor

👤 Case review

✗ Escalate / SAR



Example score to AML probability relationship

AML score	AML probability	Suggested action
140	0.4%	Approve
360	4.0%	Approve / monitor
640	19%	Case review
910	81%	Escalate / SAR

Illustrative examples – actual thresholds vary by bank, model and regulation.



How banks use the AML score



Transaction monitoring



Investigator alerts



Case prioritization



Enhanced due diligence



SAR support



Model and rule tuning



Key model governance points

- 1 Data quality matters
- 2 Use labelled case outcomes where available
- 3 Combine models with expert rules
- 4 Monitor drift and emerging laundering patterns
- 5 Check explainability and regulatory compliance
- 6 Keep human investigator oversight

Customer Churn Prediction

A core banking use case for estimating the probability that a customer will leave the bank



Business objective

- Identify customers at risk of leaving
- Support targeted retention actions
- Protect revenue, loyalty and lifetime value



LOYAL
CUSTOMERS
STRONGER
RELATIONSHIPS



Model learns patterns from historical retained and churned customers.

Typical inputs



Product
usage



Transaction
activity



Customer
tenure



Complaints /
service issues



Digital
engagement



Balance
trend



Pricing /
fees



Demographics
and segment



What is churn probability?

**Churn probability = likelihood
that a customer will leave the bank**

For each customer, the model estimates a probability between 0% and 100%.

Higher probability = higher churn risk



How the process works

DATA → INSIGHTS → BETTER DECISIONS



Customer data

Products, activity
and interactions



Data
preparation

Clean, validate
and combine data



Feature
engineering

Create behavioral
and trend features



Machine learning
algorithm

Examples: logistic
regression, random forest,
gradient boosting,
XGBoost



Churn
score

Continuous
score



Probability
mapping

Convert score to
probability of churn



Decision /
action



No action



Retention offer



Relationship
manager call



Priority save action



Example score to churn probability relationship

Churn score	Churn probability	Suggested action
160	0.6%	No action
390	5.5%	Monitor
650	24%	Retention offer
910	78%	Priority save action

Illustrative examples – actual thresholds vary by bank, model and portfolio.



How banks use the churn score



Retention
campaigns



Next best
action



RM
prioritization



Offer
personalization



Portfolio
monitoring



Customer value
management



Key model governance points

1

Data quality matters

2

Use reliable churn labels

3

Balance accuracy and actionability

4

Monitor drift and recalibrate

5

Check explainability and fairness

6

Combine model output with business judgment

Automated Ticket Routing

Using AI to send each customer request to the right team, faster



Right request.
Right expert.
A better customer experience.



Business objective

- Automatically route incoming customer tickets to the most appropriate team/agent
- Reduce resolution time and operating costs
- Improve customer satisfaction



Typical tickets

- Card blocked / limit increase
- Loan application status
- Account information
- Digital banking issues
- Complaints
- Other requests (many categories)



Why AI / ML?

- Large and growing volume of tickets
- Text is often unstructured (free text, emails, chat)
- Manual routing is slow and error-prone
- ML can understand intent, context and urgency
- Learns from historical routing and outcomes



Business value

- Faster response and resolution
- Higher first-contact resolution rate
- Better use of specialist teams
- More consistent and scalable service
- Improved customer satisfaction (NPS/CSAT)

How it works

1. Incoming ticket



2. Data preparation

- Clean and normalize text
- Detect language
- Remove personal data (PII)
- Combine with customer context (e.g. product, segment, history)

3. Text representation

- Convert ticket text to numerical features (using NLP / embeddings)
- Include additional features (e.g. customer type, channel, time of day)

Example (embedding)
[0.12, -0.03, 0.45, ...]

4. Machine learning model

- Supervised classification model trained on historical tickets and their correct department
- Examples: transformer-based model (e.g. BERT), logistic regression, SVM, gradient boosting
- Outputs probability for each possible team

5. Predicted routing

Team	Probability
Cards	0.78
Loans	0.08
Digital Banking	0.06
General Info	0.05
Complaints	0.03

6. Automated action

- Route ticket to the team with highest probability
- Optionally apply business rules (e.g. VIP customers, high-priority keywords)
- Notify agent and customer
- Continuous learning from feedback and manual corrections



Example

Customer: "I've been charged twice for the same transaction. Can you refund one of them?"

Predicted team: Payments & Transactions

Probability: 0.82

Previous similar tickets: 1,240 (82% correctly routed)



Model performance (example)

Accuracy (top-1)	87%
Accuracy (top-3)	95%
Reduction in misrouted tickets	-60%
Average handling time	-40%



Key success factors

- High-quality labeled historical data
- Good text preprocessing and NLP model
- Regular model retraining and monitoring
- Human-in-the-loop for ambiguous cases
- Integration with ticketing system
- Clear routing taxonomy and ownership



Result

- ✓ Faster and smarter service
- ✓ More efficient teams
- ✓ Happier customers
- ✓ Scalable to new products and channels

Turn every customer request into the right conversation.

People + AI for a better banking experience.

AI-Driven Marketing Campaigns

Reach the right customer with the right offer at the right time



More relevant
Conversations.
Stronger Relationships.

A tailor-made
offer for you



Business objective

- Increase customer engagement and product adoption
- Grow revenue and customer lifetime value
- Deliver personalised, relevant offers
- Optimise marketing spend (right offer, right customer, right channel)



Data sources

- Customer profile (demographics, segment, relationship value)
- Transaction history and behaviour
- Product holdings and usage patterns
- Channel interactions (web, app, branch)
- Campaign history and responses
- External data (e.g. credit bureau, geo, lifestyle signals)



Why AI / ML?

- Finds complex patterns in customer behaviour
- Predicts response probability and next best offer
- Enables real-time personalisation at scale
- Learns from past campaigns to continuously improve
- Optimises offers, channels and timing using data



Business value

- Higher response and conversion rates
- Increased product cross-sell and up-sell
- Higher customer lifetime value (CLV)
- More efficient use of marketing budget
- Improved customer satisfaction and loyalty
- Measurable and data-driven decisions

How it works

1. Customer data



Collect and integrate data from multiple sources (Core Banking, CRM, digital, external data)



Customer ID: 458732
Age: 34
Segment: Affluent
Main products: Current account, Credit card
Digital activity: High



2. Data preparation

- Clean and unify data
- Create behavioural and value-based features
- Handle missing values
- Build a single customer view

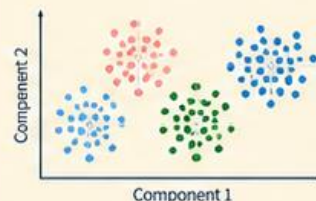
Example features

Avg. monthly balance	5,200
Card spend (3m)	2,340
Products owned	3
Digital logins (3m)	27
Tenure (years)	5



3. Customer segmentation (optional)

- Cluster customers with similar characteristics (e.g. K-means)
- Create meaningful segments (e.g. young professionals, families, high net worth)



4. Predictive model (ML)

- Train a model to predict the probability of response to each offer
- Examples: logistic regression, gradient boosting, random forest, neural networks
- Input: customer features, segment, offer, channel, timing
- Output: response probability for each customer-offer combination

$$P(\text{Response} \mid \text{customer, offer}) = 0.73$$



5. Next best offer and channel

- Select the offer with the highest expected value (e.g. response probability $\times$ expected revenue)
- Choose the best channel (e.g. app, email, SMS, branch)
- Apply business rules and constraints (e.g. eligibility, frequency caps)

Customer	Top offer	Channel
Segment A	Personal loan	App
Segment B	Premium card	Email
Segment C	Savings account	SMS



6. Campaign execution and learning

- Launch personalised campaigns
- Monitor responses and conversions
- Measure results (A/B testing)
- Feed outcomes back into the model
- Continuously retrain and improve



Learn from results to make next campaigns even better



Example

Goal: promote a premium credit card
Model predicts: 73% probability of response
Expected incremental revenue: €120
Selected channel: mobile app notification
Result: customer accepts the offer



Model performance (example)

AUC (ROC)	0.82
Precision (top 10%)	0.68
Lift (top 10% vs. random)	4.5×
Incremental revenue	+22%



Key success factors

- High-quality and integrated customer data
- Relevant and well-designed offers
- Robust model validation and monitoring
- Inclusion of business rules and risk checks
- Continuous learning from campaign results
- Alignment between marketing, data and business teams



Result

- ✓ Higher conversion rates
- ✓ Increased revenue and CLV
- ✓ More relevant customer communications
- ✓ Better use of marketing budget
- ✓ Stronger, long-term customer relationships

From data to personalised opportunities.

Banking for a brighter tomorrow.

Typical Generative AI Use Cases in a Bank

Text and Documents

- Assistant on regulations, policies and internal procedures, with source citations (RAG)
- Drafting: responses to complaints, letters to customers, credit reports, minutes (generation)
- Summarization of contracts, credit files, research reports, incoming regulatory texts (summarization)
- Structured extraction from documents: financial statements, payslips, contracts, letters of credit (JSON extraction)
- Comparison between contract versions and checking clauses against policy (guided analysis)
- Translation and language simplification for transparency towards the customer (transformation)
- Classification and routing of e-mails, PEC and tickets with rationale (zero-shot classification)

Conversation and Voice

- Customer chatbot on products and operations, with handover to the operator (RAG + guardrails)
- Real-time operator copilot: response suggestions, procedure to follow (RAG)
- Call transcription, summarization and categorization; verification of promises made to the customer (speech-to-text + summarization)
- Conversation quality analysis for coaching and compliance (evaluation with LLM)
- Voice assistant for simple operations and conversational IVR (speech-to-text, text-to-speech)

Code and IT

- Assistants for development, unit tests, code review, documentation (code generation)
- Migration from legacy languages (COBOL, PL/I) and explanation of undocumented systems (code translation)
- Generation of SQL queries and reports from natural-language questions for analysts (text-to-SQL)
- IT incident triage and post-mortem summarization (RAG, summarization)

Data

- Synthetic data for testing, development and training without exposure of personal data (tabular generation)
- Enrichment and normalization of master data and product descriptions (transformation)
- Assisted labeling of datasets for classical machine learning, validated by experts (LLM as annotator)

Training and Knowledge

- Training paths and customer-conversation simulations, quizzes generated from documentation (generation)
- Semantic search across the entire document repository with concise answers (RAG)

Images, Video and Audio

- Reading poor-quality documents and photos: cheques, identity cards, invoices, insurance damage (multimodal models)
- Generation of marketing materials and creative variants, within brand guidelines (image generation)
- Video and audio for training and internal communication, multilingual dubbing (audio/video generation)
- Defense against voice and visual deepfakes in identification processes and telephone fraud (detection) ⚠

Agents and Processes

- Verification of credit applications: document collection, checks, summary for the decision-maker, with human approval (agent with tools)
- Onboarding and KYC: collection, verification and request for missing documents (agent)
- Anti-money laundering investigations: case file, searches on open sources, draft of the report (agent + RAG)
- Preparation of commercial meetings: customer profile, latest contacts, proposals to make (agent + RAG)
- Automation of back-office flows with exceptions brought to the operator (agent with human-in-the-loop)

Banking Customer Chatbot

Reliable answers on products and operations, with handover to a human operator

Generative AI with Retrieval-Augmented Generation (RAG) and guardrails, integrated with live support.



Hi!
How can I help you today?

Get answers.
Solve issues.
When needed,
we connect you
with a specialist.



Business objective

- Provide fast, accurate and consistent answers to customer questions
- Reduce contact center workload and costs
- Improve customer satisfaction (24/7 support)
- Ensure compliance and risk control
- Seamless handover to a human operator when needed



Typical customer questions

- “What are the fees for a current account?”
- “How can I increase my credit card limit?”
- “Where can I find my account statement?”
- “What documents are needed for a mortgage?”
- “How long does a bank transfer take?”
- “I have a problem with a transaction ...”



Why Generative AI (with RAG)?

- Understands natural language
- Retrieves answers from trusted bank content
- Provides clear, conversational responses
- Can handle complex, multi-step questions
- Reduces manual search for information
- Learns from usage (with human feedback)



Business value

- Higher customer satisfaction (CSAT)
- Lower contact center volume (deflection)
- Faster and more consistent responses
- Better use of human agents (focus on complex cases)
- Scalable to more products, languages and channels
- Measurable ROI

How it works

1. Customer asks a question

- Web chat (website)
- Mobile banking app
- Voice (optional)
- Natural language question

“What are the fees for an international transfer?”

2. Retrieve relevant information (RAG)



Bank knowledge base

- Product information
- Policies and procedures
- FAQs
- Regulatory documents
- Internal operational guides



Semantic search

(find the most relevant content)

3. Generate a grounded answer



Large Language Model (with retrieved context)

- Uses the retrieved information to generate a clear and accurate answer
- Cites sources when appropriate
- Follows bank's tone and style

“You can make an international transfer for €5 via our app...”
(with link to details)



4. Guardrails & safety checks

- ✓ Use only trusted sources (RAG)
- ✓ Filter out-of-scope or sensitive topics
- ✓ Detect and block harmful content
- ✓ Ensure compliance with regulatory rules
- ✓ Check for factual consistency
- ✓ Add disclaimers when needed



5. Handover to a human (operator)

If the bot cannot answer or the customer requests it:

- Seamless transfer to a live agent
- Provide the conversation summary and context
- Customer does not need to repeat the question
- Agent continues the conversation in the same channel



6. Continuous improvement

- Monitor conversations and feedback
- Identify gaps in knowledge base
- Measure accuracy, deflection rate and CSAT
- Use human feedback to improve responses
- Update content and guardrails regularly



A smarter chatbot over time



Example conversation



How can I increase my credit card limit?



You can request a higher credit card limit directly from the app. Go to Cards > Manage limit and follow the steps. The request is subject to a credit assessment. Here are the details: [\[link\]](#)



Great, thanks!



Key metrics (example)

Answer accuracy	92%
Contact deflection rate	55%
Customer satisfaction (CSAT)	4.6 / 5
Average response time	2 seconds
Handover rate to human	15%



Success factors

- ✓ High-quality, up-to-date content
- ✓ Effective retrieval and context management
- ✓ Clear guardrails and risk controls
- ✓ Seamless integration with contact center tools
- ✓ Continuous monitoring and human feedback
- ✓ Focus on user experience and trust



Result

- ✓ Customers get reliable answers, anytime, anywhere
- ✓ Human agents focus on higher-value cases
- ✓ Lower operational costs
- ✓ Stronger trust and compliance
- ✓ Scalable, multilingual support
- ✓ Measurable business impact

AI that helps your customers move forward.

People + AI for a better banking experience.

Structured Extraction from Documents with GenAI

Turn unstructured documents into structured data (JSON)

Financial statements, payslips, contracts, letters of credit and more.



From documents
to data.
Faster, more accurate,
more value.



Business objective

- Automatically extract key information from banking documents
- Reduce manual data entry and processing time
- Improve accuracy and consistency
- Enable straight-through processing (STP)
- Free staff for higher-value tasks



Typical documents

- Financial statements (P&L, balance sheet, cash flow)
- Payslips (salary, employer, deductions)
- Contracts (loan agreements, customer contracts)
- Letters of credit (parties, amounts, dates, terms)
- Other documents (invoices, IDs, tax forms, ...)



Why Generative AI?

- Understands natural language and document layout (text, tables, forms)
- Can extract complex information across pages
- Works with diverse formats and templates
- Reduces the need for rigid rule-based systems
- Produces structured output (e.g. JSON)
- Can handle variations in language and formatting



Business value

- Faster turnaround time (hours → minutes)
- Higher accuracy and fewer errors
- Scalable processing of large document volumes
- Improved compliance and auditability
- Better customer and employee experience
- Measurable cost savings

How it works

1. Document input

- PDF
- Images (scans, photos)
- Word, Excel
- Emails (attachments)
- ...

Multiple formats,
multiple templates,
multiple languages.



2. Preprocessing

- Convert to text (OCR if needed)
- Clean and normalize content
- Detect document type
- Split into sections (e.g. tables, clauses, headers)

INCOME STATEMENT	
Revenue	1,250,000
Cost of goods sold	(720,000)
Gross profit	530,000
Operating expenses	(320,000)
Operating profit	210,000



3. Information extraction (with GenAI)

- Use an LLM with instructions and examples (few-shot)
- Extract relevant fields and understand their meaning
- Handle tables, key-value pairs and unstructured text
- Apply validation and business rules

Prompt + document content
→ structured JSON



4. Validation and guardrails

- Check data types, formats and business rules
- Validate amounts, dates, names (e.g. with regex or reference lists)
- Detect and flag low-confidence extractions
- Human review for exceptions or critical cases
- Log prompts, outputs and changes for auditability



5. Structured output (JSON)

```
{
  "document_type": "financial_statement",
  "company_name": "ABC S.p.A.",
  "period_end": "2024-12-31",
  "currency": "EUR",
  "revenue": 1250000,
  "gross_profit": 530000,
  "net_profit": 120000,
  "total_assets": 3700000,
  "total_liabilities": 2100000,
  "extracted_fields_confidence": {
    "revenue": 0.98,
    "net_profit": 0.95
  }
}
```



6. Integration and usage

- Send structured data to banking systems (e.g. loan origination, KYC, compliance)
- Trigger automated workflows
- Store data and extraction metadata
- Monitor performance and continuously improve

From documents to
trusted data, ready for action.



Examples by document type

Financial statements

- Company name
- Reporting period
- Revenue, EBITDA
- Total assets/liabilities
- Key ratios

Payslips

- Employee name
- Employer
- Gross and net salary
- Deductions
- Employment type

Contracts

- Parties
- Effective date
- Contract value
- Key clauses
- Termination terms

Letters of credit

- Applicant and beneficiary
- Issuing bank
- Amount and currency
- Expiry date
- Applicable rules (e.g. UCP 600)



Performance (example)

Field extraction accuracy	95%
Document classification	98%
End-to-end processing time	-80%
Manual data entry effort	-70%



Result

- ✓ Structured, reliable data (JSON)
- ✓ Faster processing and decisioning
- ✓ Higher accuracy and consistency
- ✓ Scalable to multiple document types
- ✓ Audit trail and compliance
- ✓ More efficient and satisfied teams

Unlock the value of your documents with Generative AI.

Trusted data. Stronger decisions. A better banking experience.

Reading Poor-Quality Documents and Photos with AI

Extract accurate information from real-world images, even when quality is low
Cheques, identity cards, invoices, insurance damage and more, using multimodal models (vision + language).



Any document.
Any format.
Any quality.
Useful information.

Business objective

- Automatically read and extract key information from poor-quality documents and photos
- Reduce manual data entry and processing time
- Speed up customer onboarding and claims handling
- Improve accuracy and consistency
- Enhance customer and employee experience

Typical documents and photos

- Cheques (handwritten or scanned)
- Identity cards, passports, driving licences
- Invoices and receipts (paper, mobile photo)
- Insurance damage photos (vehicles, property)
- Contracts and application forms
- Other documents (statements, letters, etc.)

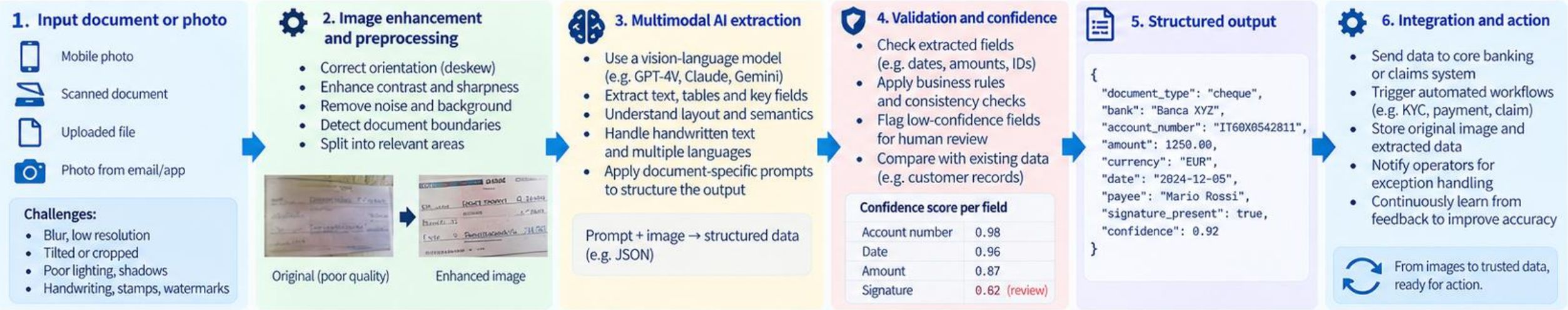
Why multimodal AI?

- Understands both text and images
- Can handle low-quality, blurry, tilted or partially occluded documents
- Recognises handwritten and printed text (OCR)
- Understands document layout and context
- Extracts structured data (e.g. JSON)
- More robust than traditional OCR alone

Business value

- Faster processing (hours → seconds)
- Higher extraction accuracy, even on poor quality
- Scalable across multiple document types
- Reduced operational costs
- Supports straight-through processing (STP)
- Better customer service (faster onboarding and claims)

How it works



Cheques

Identity cards

Invoices

Insurance damage

Performance (example)

Field extraction accuracy	95%
Handwritten text accuracy	90%
End-to-end processing time	5–10 sec
Reduction in manual effort	-80%

Result

- ✓ Accurate data from real-world documents
- ✓ Faster onboarding, payments and claims
- ✓ Less manual work and fewer errors
- ✓ Scalable to multiple document types and languages
- ✓ Higher customer and employee satisfaction
- ✓ Measurable cost savings

Video and Audio for Training and Internal Communication

Create engaging, multilingual content with AI

Turn ideas into professional videos and audio, with multilingual dubbing, in a fraction of the time.



One message.
Many languages.
A more connected bank.

Inform
Train
Engage
Everywhere.



Business objective

- Create high-quality training and internal communication content quickly
- Reach employees in multiple languages
- Reduce production time and costs
- Ensure consistent and up-to-date messaging
- Increase engagement and knowledge retention



Typical use cases

- Training videos (compliance, products, processes)
- CEO and leadership messages
- Internal announcements and change programs
- Micro-learning clips and tutorials
- Multilingual dubbing and subtitles
- Audio podcasts and voiceovers
- Other formats (e.g. animations, avatars)



Why Generative AI?

- Generates realistic video, audio and avatars
- Automates dubbing and translation at scale
- Maintains voice consistency across languages
- Adapts content to different audiences and formats
- Reduces dependence on external production
- Enables frequent updates and personalization



Business value

- Faster time to market (days → hours)
- Lower production costs
- Broader reach and inclusivity (multilingual)
- More engaging and effective learning
- Consistent brand and regulatory messaging
- Measurable impact on knowledge and behavior

How it works

1. Content and objectives



- Define the goal (training, information, announcement)
- Prepare key messages and script (or use an LLM to draft it)
- Choose the target audience and languages
- Select format (video, audio, avatar, animation)

“Create a 3-minute training video on anti-money laundering for our international teams.”



2. Generate the content

- Create video with AI (real avatar, synthetic avatar or animation)
- Generate voiceover in the source language (text-to-speech)
- Add visuals, slides, and graphics
- Edit and refine



3. Multilingual dubbing and localization

- Automatically translate the script
- Generate natural-sounding dubbing in multiple languages
- Add subtitles (optional)
- Adapt visuals or on-screen text
- Maintain consistent tone and brand



4. Review and quality check

- Check accuracy of translation and terminology
- Verify audio/video quality and lip-sync (if avatar)
- Ensure compliance with internal and regulatory guidelines
- Get feedback from stakeholders
- Make final adjustments



5. Publish and distribute

- Export in the required formats (e.g. MP4, audio, subtitles)
- Publish on internal channels (intranet, LMS, Teams, etc.)
- Track engagement and completion rates
- Update easily when content changes



6. Measure and improve

- Monitor views, completion rates and feedback
- Assess learning impact (e.g. quiz results)
- Identify content to update or localize in new languages
- Continuously improve based on data and user feedback



Keep your content relevant and effective over time.



Example: compliance training video (multilingual)



- English – Original
- French – Doublage IA
- German – KI-Synchronisation
- Italian – Doppiaggio IA
- Spanish – Doblaje con IA



Typical results (example)

Production time	-70%
Production cost	-60%
Employee reach	3×
Completion rate	+40%
Knowledge retention	+30%



Key success factors

- ✓ Clear objectives and target audience
- ✓ High-quality and accurate translation
- ✓ Realistic and natural-sounding voices
- ✓ Consistent tone and branding
- ✓ Review and human oversight
- ✓ Easy distribution and tracking
- ✓ Regular updates and continuous improvement



Result

- ✓ Engaged and better-informed employees
- ✓ Consistent global communication
- ✓ Faster and more efficient content production
- ✓ Scalable to multiple languages and formats
- ✓ Stronger compliance and risk awareness
- ✓ Measurable impact on learning and behavior

The right message. In every language. For a stronger bank.

People + AI for a more inclusive and effective bank.

Defending Against Voice and Visual Deepfakes

Protecting identification processes and preventing telephone fraud

Detect AI-generated or manipulated voices and images to keep your customers and your bank safe.



Real customers.
Real security.
A more trusted bank.

Detect
Prevent
Protect



Business objective

- Prevent identity fraud using AI-generated voices and images
- Protect customers and the bank from financial and reputational losses
- Ensure secure and trusted identification processes (digital and phone)
- Comply with regulatory requirements (e.g. KYC, AML, consumer protection)



Typical fraud scenarios

- Impersonation in phone calls (voice cloning)
- Fake identity documents (photos/videos)
- Deepfake videos during remote onboarding
- Synthetic voices to authorise transactions
- Social engineering with AI-generated content
- Combination of deepfakes with stolen data
- Other channels (video calls, chat, messaging)



Why AI is needed

- Deepfakes are increasingly realistic and easy to produce
- Humans struggle to detect them reliably
- AI can analyse subtle signals in voice, image and video
- Combines multiple checks (biometric, behavioural, contextual)
- Works at scale and in real time
- Continuously learns from new fraud patterns



Business value

- Reduced fraud losses
- Higher security and customer trust
- Faster and safer customer onboarding
- Better protection for vulnerable customers
- Compliance with regulatory expectations
- Lower operational costs (fewer manual checks)
- Measurable reduction in successful fraud attempts

How it works

1. Capture and verify



Voice: phone call or voice recording



Image/video: ID document, selfie or video call

Collect media and contextual data (device, location, behaviour).



2. Analyse with AI models

- Voice analysis: detect synthetic speech and cloning
- Face analysis: detect manipulated or AI-generated images/videos
- Liveness detection (e.g. blink, head movement)
- Behavioural and contextual signals (device, network, pattern)



3. Combine and assess risk

- Aggregate multiple signals (voice, face, behaviour, device)
- Compute a fraud probability score
- Apply business rules and risk thresholds



4. Action

- Low risk: proceed automatically
- Medium risk: additional verification (e.g. OTP, live check)
- High risk: block and alert fraud team
- Log decision and evidence for audit

Explainable decision:
key factors and evidence



5. Continuous learning

- Use feedback from confirmed fraud cases
- Update models with new deepfake techniques
- Monitor performance and drift
- Share intelligence across channels and entities



Adapt to new threats.
Stay one step ahead.



6. Integration

- Integrate with KYC, onboarding and transaction systems
- Real-time APIs for voice and image checks
- Case management for alerts
- Dashboards and reporting for risk and compliance



Seamless integration
into existing processes.



Examples of deepfakes

Real vs AI-generated face



Real vs cloned voice



Manipulated ID document



Detection performance (example)

Deepfake face detection (image)	96%
Deepfake video detection	94%
Synthetic voice detection	92%
Overall accuracy	93%

Performance depends on data quality, attack type and operating conditions.



Result

- ✓ Significant reduction in successful fraud
- ✓ More secure and trusted identification
- ✓ Better customer experience
- ✓ Compliance with regulatory requirements
- ✓ Scalable and real-time protection
- ✓ Measurable business impact

Trust is hard to fake.

People + AI for a safer banking experience.

AI Agent for Credit Application Verification

From documents to a clear recommendation, with human approval

Collect. Check. Analyse. Summarise. Support the decision — with a human in control.



AI does the heavy lifting.
People make the decision.



Faster
safer
fairer
lending



Business objective

- Speed up and improve the accuracy of credit application review
- Ensure complete and compliant documentation
- Support consistent and data-driven decisions
- Reduce operational costs and turnaround time
- Keep a human in the loop for final approval



Typical documents

- Identity documents (ID, passport)
- Proof of income (payslips, tax returns)
- Bank statements
- Employment contract
- Loan purpose and supporting documents (e.g. invoicing, quotes)
- Additional documents (e.g. existing loans, collateral, guarantees)



Why an AI agent?

- Understands and extracts information from multiple document types
- Uses tools to perform automated checks (internal and external)
- Cross-validates data and detects inconsistencies
- Produces a structured summary and risk assessment
- Saves analysts' time on repetitive tasks
- Improves consistency and auditability



Business value

- Faster time to decision (hours → minutes)
- Higher data quality and fewer errors
- Better risk assessment and fraud detection
- Improved customer experience
- More consistent and explainable decisions
- Scalable to higher volumes
- Regulatory compliance and full audit trail

How it works

1. Submit application



Customer submits application (via web or branch)



Uploads required documents (or provides consent to retrieve)



Application is created in the bank's system

Guided checklist to ensure all required documents are provided.



2. Extract and validate data

- Extract key information from documents (OCR + LLM)
- Validate data and check for completeness
- Detect inconsistencies (e.g. income vs. bank statements)
- Classify document types and assess authenticity



3. Run automated checks (using tools)

- Check identity (KYC, sanctions, PEP)
- Verify income and employment (with internal and external sources)
- Check credit history (internal/external)
- Assess existing exposures and limits
- Evaluate collateral (if any)
- Apply credit policy rules



4. Generate summary and recommendation

- Create a structured summary of key information
- Highlight risks, inconsistencies and missing items
- Provide an explainable recommendation (e.g. approve, conditionally approve, request more information, reject)
- Include links to source documents and checks performed

Clear, concise and auditable output for the decision-maker.



5. Human review and decision

- Credit analyst reviews the AI-generated summary
- Checks flagged items and, if needed, requests additional information
- Approves, rejects or asks for changes
- Decision is recorded in the system with full audit trail

Approve Approve with conditions Reject



6. Notify and complete

- Customer is notified about the decision
- If approved, the loan onboarding process continues
- If more information is needed, the customer is contacted with clear instructions
- All data and interactions are stored for compliance and future reference

Continuous learning from feedback and outcomes.



Example output (summary for the decision-maker)

Application summary

Applicant	Mario Rossi
Requested amount	€ 50,000
Loan purpose	Home renovation
Employment	Permanent, 3 years
Monthly income (verified)	€ 2,800
Existing loans	€ 12,000
Credit score	720 (good)

Risk assessment

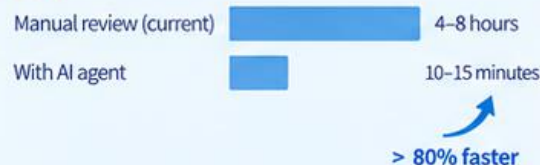
Low risk

- All required documents provided
- Identity verified (no alerts)
- Income consistent with bank statements
- No negative credit events
- Debt-to-income ratio: 28% (within policy)

Recommendation: Approve



Typical time and effort (example)



Key success factors

- ✓ High-quality document extraction
- ✓ Reliable data sources and integrations
- ✓ Clear business rules and risk policies
- ✓ Human oversight and accountability
- ✓ Explainable and transparent outputs
- ✓ Continuous monitoring and improvement
- ✓ Compliance with regulatory requirements



Result

- ✓ Faster and more accurate decisions
- ✓ Reduced operational costs
- ✓ Improved risk management
- ✓ Better customer experience
- ✓ Consistent and compliant process
- ✓ Scalable to higher volumes
- ✓ Measurable business impact

Augmenting expertise. Enabling better credit decisions.

People + AI for a more trusted and inclusive bank.

Mit: 95% of GenAI business projects do not produce value, only 5% bring returns

Unofficial AI use among employees exceeds licences, influencing job change more than official projects. This is the finding of the report 'State of AI in Business 2025', compiled by the research team Mit Nanda (Networked Agents and Decentralised AI) on more than 300 AI initiatives, 52 company interviews and 153 questionnaires collected at industry conferences

20 August 2025

Despite investments of between USD 30 and 40 billion by companies, 95 per cent of **generative artificial intelligence (GenAI)** projects do not produce any concrete impact on company accounts. This is the finding of the report '**State of AI in Business 2025**', compiled by the **Mit Nanda** (Networked Agents and Decentralised AI) research team on more than 300 AI initiatives, 52 company interviews and 153 questionnaires collected at industry conferences.

The research highlights a clear divide, dubbed the 'GenAI Divide': only five per cent of integrated pilot projects generate value, with returns in the millions, while the vast majority remain stuck at the starting line, with no effect on company accounts. It is not 'model quality' or 'regulation' that are the determining factors, according to the research, but the approach taken by companies.

Reduced testing

Popular tools such as ChatGPT and Copilot have exceeded 80% adoption and nearly 40% deployment, but they only affect individual productivity, not operating margin. Customised enterprise systems, on the other hand, are often rejected: 60% of companies evaluate them, **but only 20% get to test them** and barely 5% take them into production. The main reasons are the difficulty of integrating them into workflows, the lack of contextual learning and the lack of adaptability.

The phenomenon is reflected at sector level: out of nine sectors analysed, only technology and media show signs of structural transformation. **In healthcare, finance, retail and manufacturing, adoption remains superficial, with experiments without any particular effect.** Concrete savings emerge mainly from the reduction of external service costs: between USD 2 and 10 million per year in customer service and document processing and up to 30 per cent less costs for creative agencies.

An executive of a medium-sized manufacturing company summed it up like this: 'The enthusiasm on LinkedIn says that everything has changed, but in our operations nothing is really different. We process some contracts faster, but that's about it'.

MIT NANDA STUDY

AI in business: a flop? Success is possible — here's how

According to the report The GenAI Divide: State of AI in Business 2025, produced by MIT (Project NANDA), only 5% of artificial-intelligence projects bring benefits to companies. Read carefully, however, and the issue is not technological: it is about making good choices and building the right culture.

Published 25 Aug 2025

Maurizio Carmignani

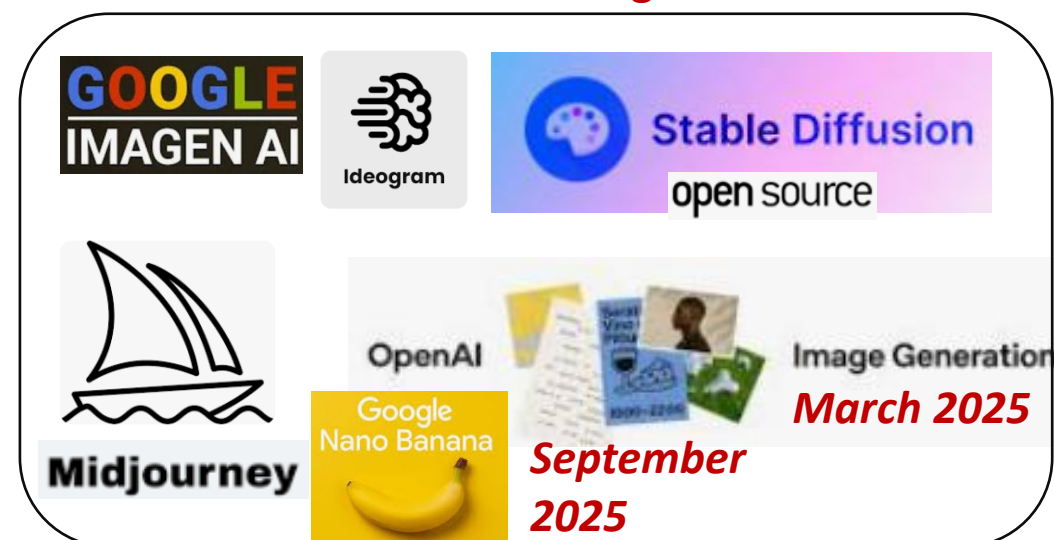
Founder & CEO – Management Consultant, Trainer & Startup Advisor

link to the
article

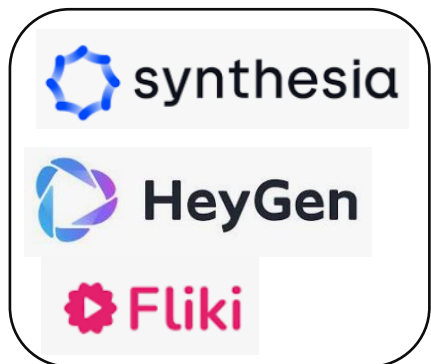
Text 2 Text (LLM)



Text 2 Image



Text 2 Video (with editor)



Text 2 Video (few seconds)

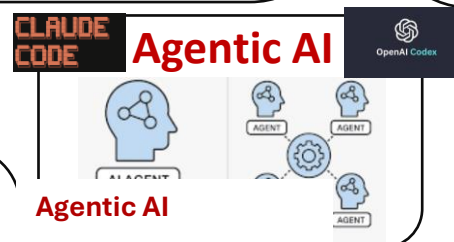
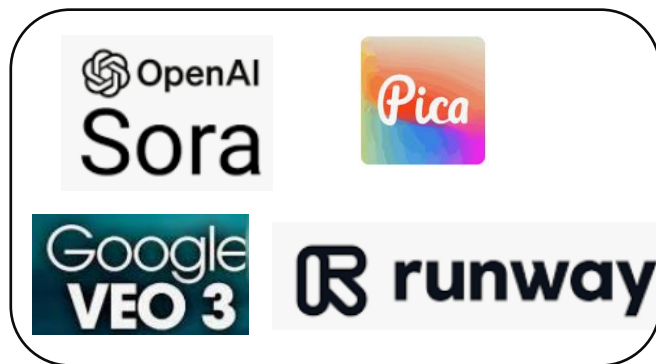
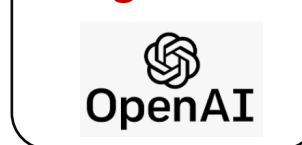


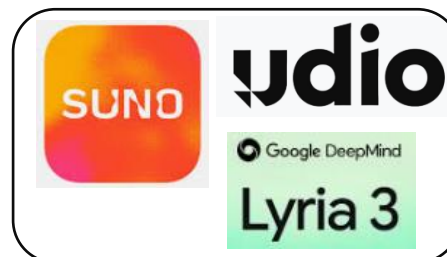
Image 2 Text



Text 2 Speech / Speech 2 Text / Speech 2 Speech



Text 2 Music



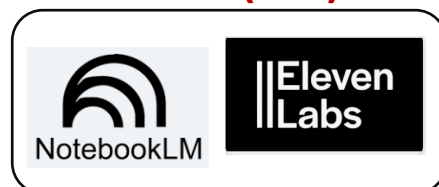
AI-enabled web search



Language Translation (T2T)



Podcast (T2S)



Text 2 Code / Code 2 Text / Code 2 Code



Real-time Language

Real-time Language Translation?

not mature yet

Fundamental distinction (often confused)

Term	Actual meaning
Classic open source	code + training data + pipeline
Open weights	weights available, training not reproducible
Closed	API only

6 Ways to Access GenAI

Common access modes for AI applications

1



Cloud tools (URL)

For end users. Usually accessed through a web interface and often offered with flat-rate subscriptions.

2



Playgrounds

For technical users. Usually offer only a few modes, mainly text and code. Pay per use (typically low cost).

Examples: OpenAI Playground, Claude Console, Google AI Studio

3



APIs

Access from Python, Java, and other programming languages. Pay per use (typically low cost).

4



Agentic AI

AI agents running in the background and performing specific tasks.

Examples: OpenAI Codex, n8n, Zapier

5



Desktop apps

Installed applications for desktop use.

Examples: ChatGPT, Claude

6



Local models

Run on a local PC or server with a GPU, using local runtimes.

Examples: Ollama



Different access modes serve different users: end users, technical users, developers, workflow automation, and local deployment.

Key AI Trends

Practical directions shaping current AI applications.



X2X transformation pipelines

Converting text, audio, images, video, and structured data from one format to another.



RAG on local data

Retrieval-augmented generation grounded on internal documents and private knowledge bases.



Fine-tuning on local data

Adapting models with organization-specific examples and domain data.



Anonymization

Removing or masking personal and sensitive information before AI processing.



Agents

AI systems that plan actions, use tools, and execute tasks in the background.



Local GPU models (Ollama and others)

Running models on local machines or servers for privacy, control, and speed.

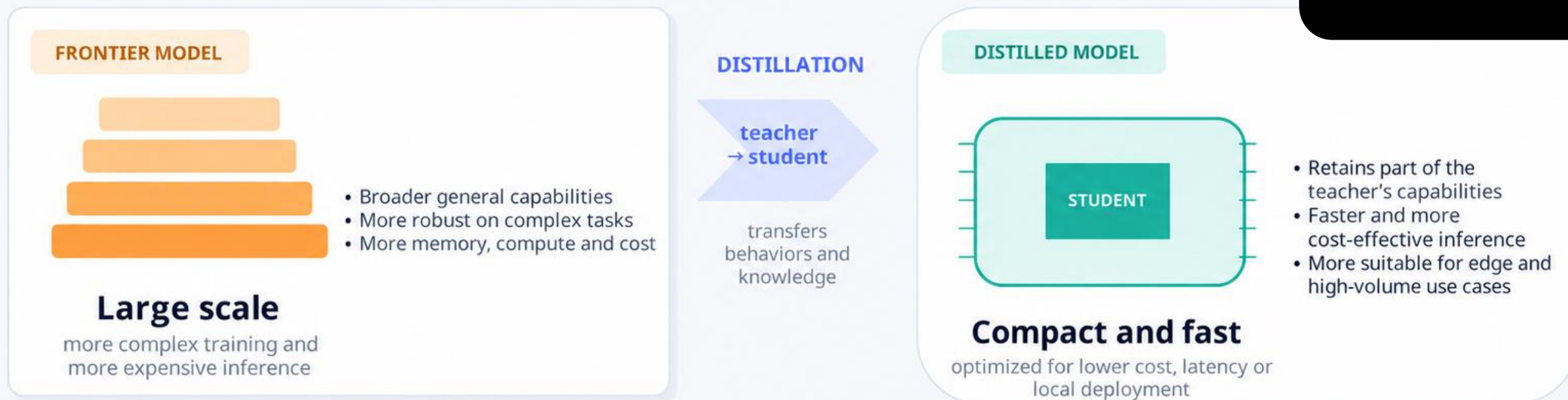


Common themes: privacy, local deployment, automation, and stronger use of proprietary data.

Frontier Model vs Distilled Model

Same family of ideas, different goals: maximum capability vs operational efficiency

[link to an MLM article on model distillation](#)



Operational comparison

CRITERION	FRONTIER	DISTILLED
Capability	broader / more extensive	more selective / compressed
Size	very large	smaller
Latency	higher	lower
Cost per use	higher	lower
Deployment	cloud / strong infrastructure	lightweight cloud, edge, on-device

1980

1990

2000

2010

2020



PERSONAL COMPUTING



MACHINE LEARNING



WWW



BROADBAND INTERNET



CLOUD COMPUTING



MOBILE DEVICE REVOLUTION

Paradigm Shifts
in Tech History



GENERATIVE AI

Centrality of Images in Generative AI



images



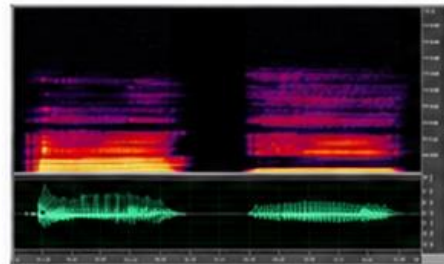
video



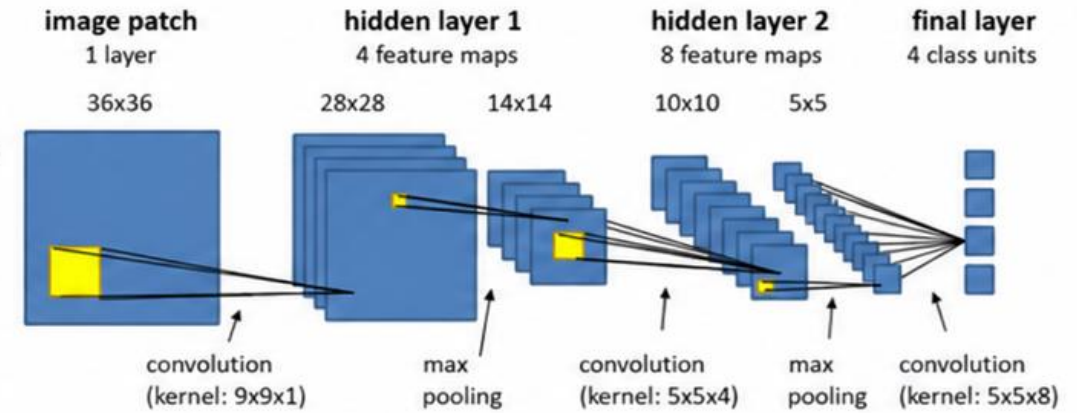
frames (images)



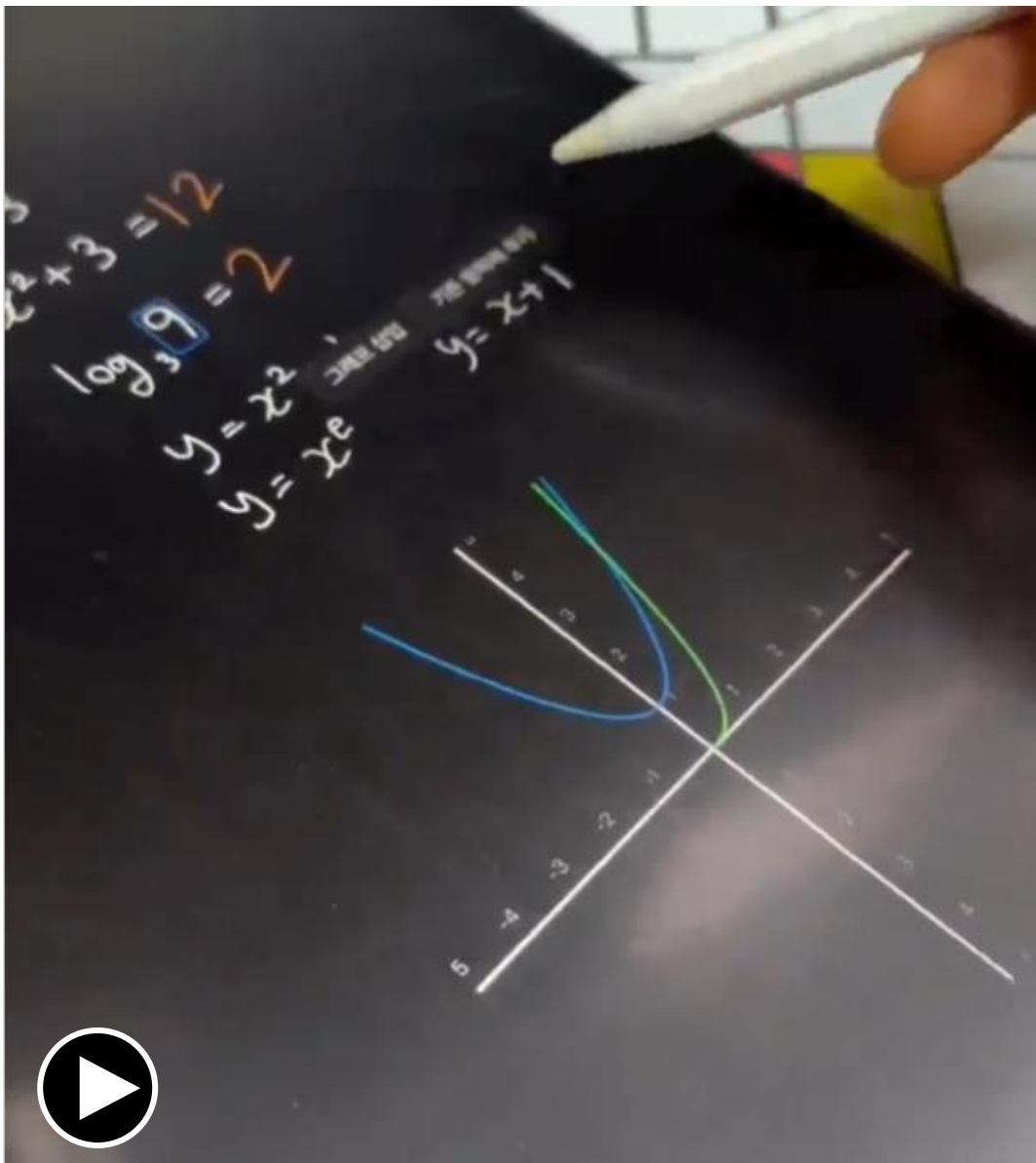
audio



spectrogram (image)



CNN for image classification



The AI-Powered Calculator app on iOS 18

A potentially misleading expression: "AI-Powered Calculator" is a marketing label that hides very different paradigms. June 2024

If by "AI-Powered Calculator app on iOS 18" you mean Math Notes in the Calculator app (the Apple Intelligence feature where you handwrite an expression and it solves it, handles variables, draws graphs, etc.), almost none of what it does is GenAI in the strict sense.

Let us break down the pipeline, because it is useful to distinguish three layers: Handwriting ("ink") recognition is perceptual/discriminative AI — computer vision with on-device neural networks mapping strokes → mathematical symbols. It is classification, not generation.

The actual solution of the expression (including variables, equations and graphs) is handled by a deterministic symbolic/CAS engine. There is no ML here at all: it is exact algebraic computation. An LLM, by contrast, would predict the result probabilistically — precisely what Apple avoided, because a generative model that "guesses" 7×8 would be unacceptable in a calculator.

The only plausibly generative part is rendering the result in your handwriting: the system generates glyphs that imitate your writing to complete the handwritten equation. That is conditioned image/ink generation — modest and localized, not a large language model or a large-scale diffusion model.

So, if it must be placed in a GenAI taxonomy, the most accurate label is "conditioned ink/image generation" limited to handwriting rendering. Everything else is hybrid AI: perception (vision) + deterministic symbolic computation. Calling it simply "generative AI" would be inaccurate, and it is exactly the kind of marketing-vs-reality conflation worth showing in class: "AI-powered" does not imply "generative", and "generative" does not imply "LLM".

If instead you mean one of the many third-party "AI Calculator" apps in the App Store, the situation changes: many of them send a photo of the problem to a VLM/LLM (such as GPT-4o), which generates steps and a natural-language explanation. Those clearly fall under multimodal GenAI (vision-language), with the associated risks of numerical hallucination.

Would you like a side-by-side slide comparing hybrid Math Notes with an LLM-based app for the course?

2. How LLMs work

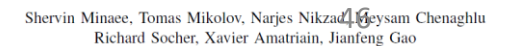


Language model

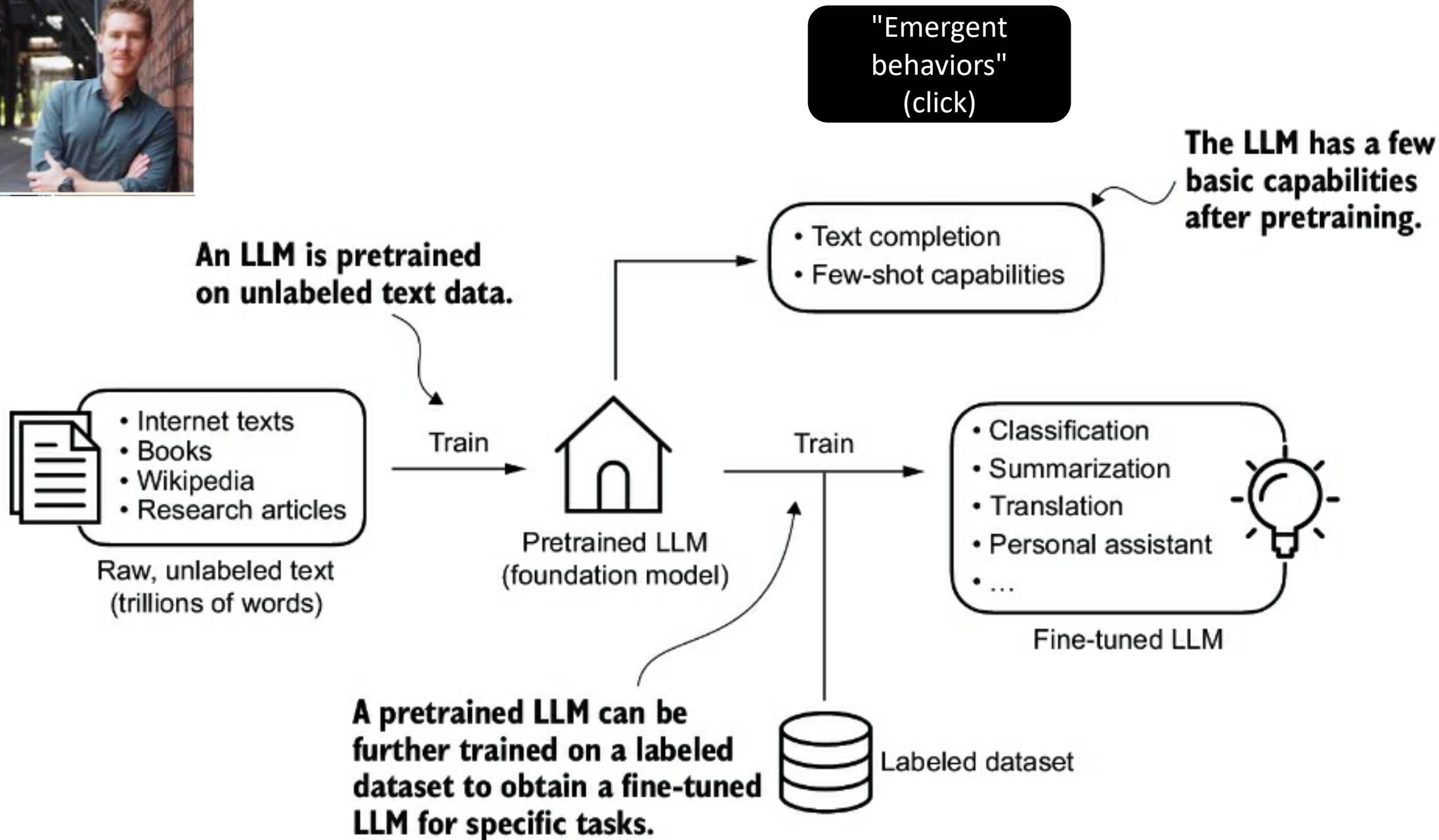
A **language model** is a [probability distribution](#) over sequences of words.^[1] Given any sequence of words of length m , a language model assigns a probability $P(w_1, \dots, w_m)$ to the whole sequence. Language models generate probabilities by training on [text corpora](#) in one or many languages. Given that languages can be used to express an infinite variety of valid sentences (the property of [digital infinity](#)), language modeling faces the problem of assigning non-zero probabilities to linguistically valid sequences that may never be encountered in the training data. Several modelling approaches have been designed to surmount this problem, such as applying the [Markov assumption](#) or using neural architectures such as [recurrent neural networks](#) or [transformers](#).

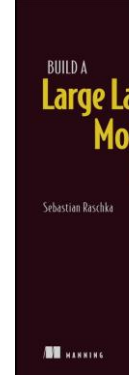
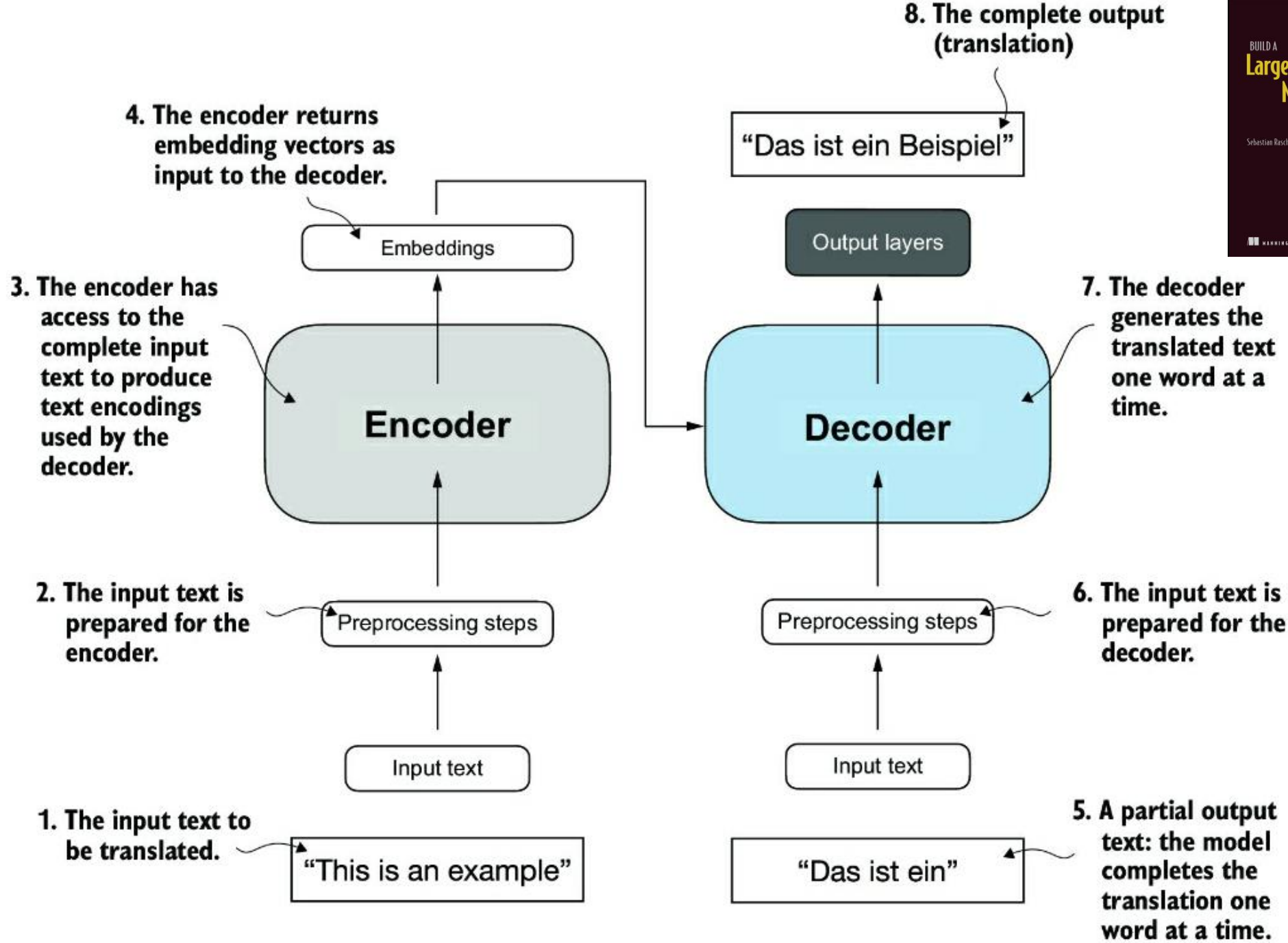
Language models are useful for a variety of problems in [computational linguistics](#); from initial applications in [speech recognition](#)^[2] to ensure nonsensical (i.e. low-probability) word sequences are not predicted, to wider use in [machine translation](#)^[3] (e.g. scoring candidate translations), [natural language generation](#) (generating more human-like text), [part-of-speech tagging](#), [parsing](#),^[3] [Optical Character Recognition](#), [handwriting recognition](#),^[4] [grammar induction](#),^[5] [information retrieval](#),^{[6][7]} and other applications.

Large Language Models: A Survey



16





"Attention Is All You Need"



The Transformer architecture – Google, 2017



Table 1.1 The pretraining dataset of the popular GPT-3 LLM

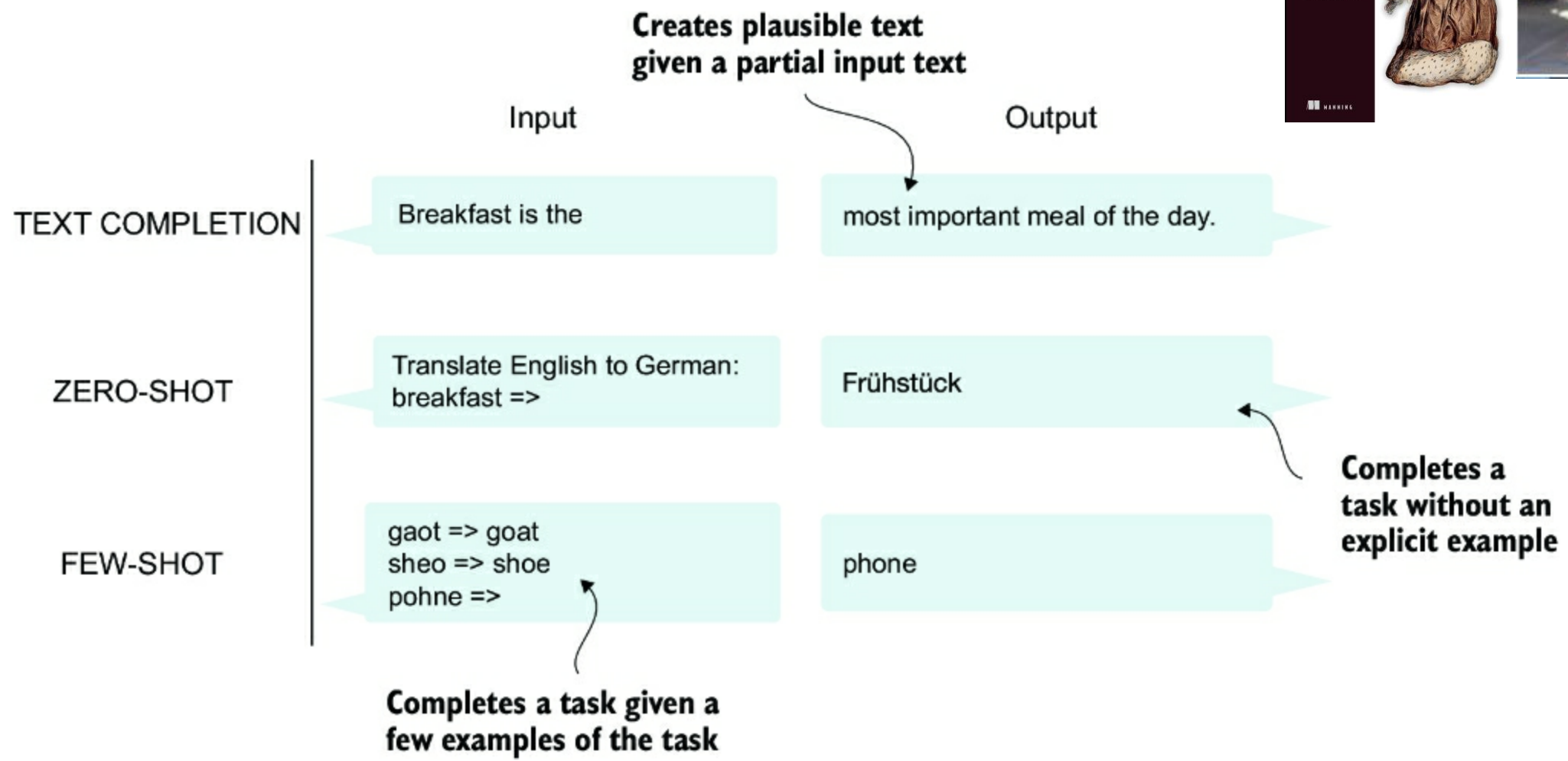
Dataset name	Dataset description	Number of tokens	Proportion in training data
CommonCrawl (filtered)	Web crawl data	410 billion	60%
WebText2	Web crawl data	19 billion	22%
Books1	Internet-based book corpus	12 billion	8%
Books2	Internet-based book corpus	55 billion	8%
Wikipedia	High-quality text	3 billion	3%

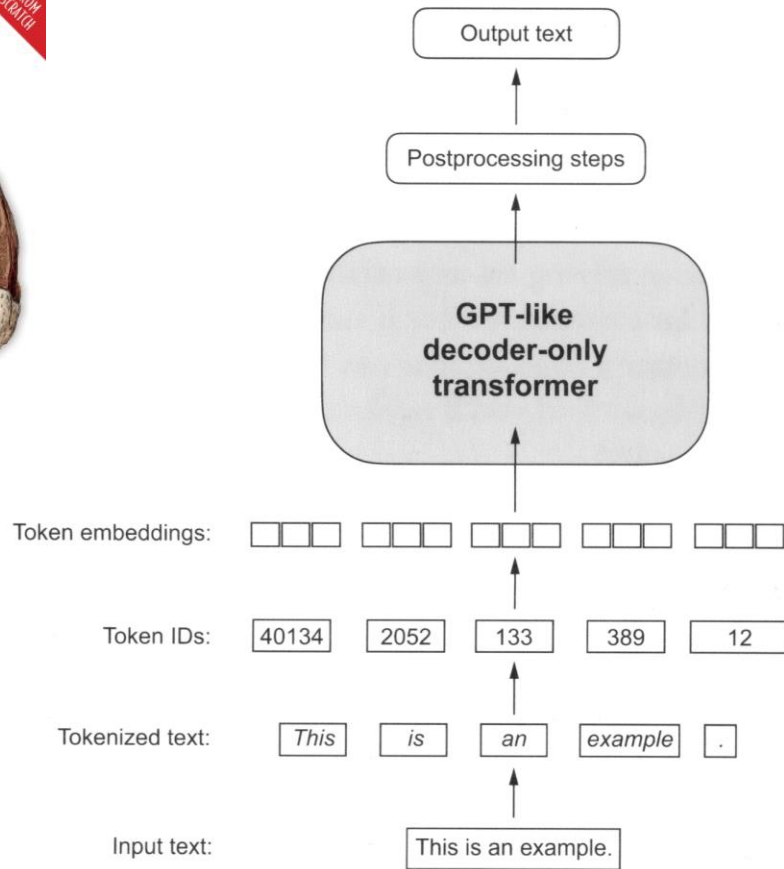
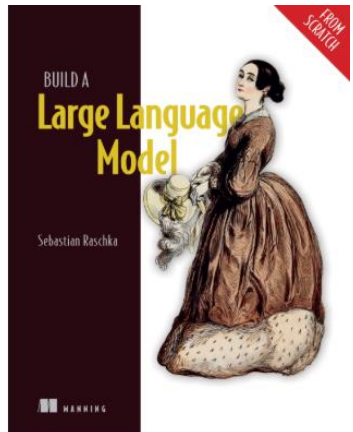
GPT-3 DATASET DETAILS

Table 1.1 displays the dataset used for GPT-3. The proportions column in the table sums up to 100% of the sampled data, adjusted for rounding errors. Although the subsets in the Number of Tokens column total 499 billion, the model was trained on only 300 billion tokens. The authors of the GPT-3 paper did not specify why the model was not trained on all 499 billion tokens.

For context, consider the size of the CommonCrawl dataset, which alone consists of 410 billion tokens and requires about 570 GB of storage. In comparison, later iterations of models like GPT-3, such as Meta’s LLaMA, have expanded their training scope to include additional data sources like Arxiv research papers (92 GB) and StackExchange’s code-related Q&As (78 GB).

The authors of the GPT-3 paper did not share the training dataset, but a comparable dataset that is publicly available is *Dolma: An Open Corpus of Three Trillion Tokens for LLM Pretraining Research* by Soldaini et al. 2024 (<https://arxiv.org/abs/2402.00159>). However, the collection may contain copyrighted works, and the exact usage terms may depend on the intended use case and country.





A view of the text processing steps in the context of an LLM. The process starts with input text, which is broken down into tokens and then converted into numerical token IDs. These IDs are linked to token embeddings that serve as the input for the GPT model. The model processes these embeddings and generates output text. Finally, the output undergoes postprocessing steps to produce the final text. This flow illustrates the basic operations of tokenization, embedding, transformation, and postprocessing in a GPT model that is implemented from the ground up in this book.

Semplice (a parte i «pericoli»)



The foundation of everything: conditional probabilities

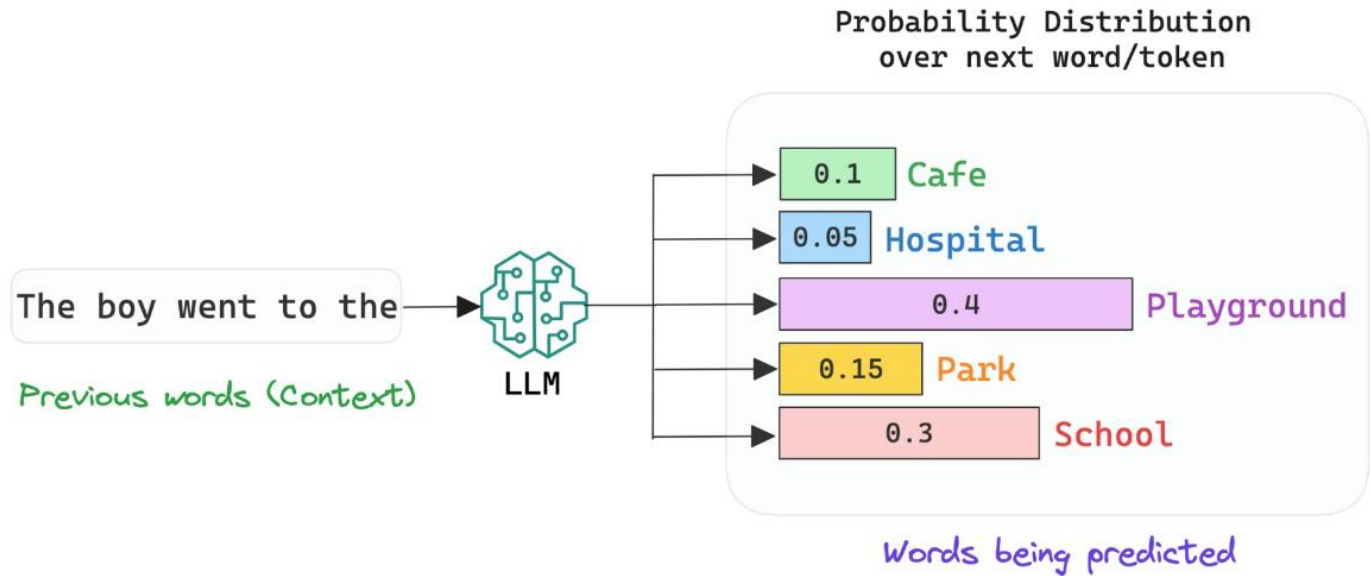
An LLM is a token classifier! (Damien)

The model underlying an LLM is a next-token predictor. At its core, a language model does one thing: given an input text (the context), it produces a probability distribution over the next token. It is not a classifier with only a few classes: it is a classifier with tens of thousands of "classes", one for each token in the vocabulary. It uses a softmax function with roughly 50,000 outputs (one per token). The classes are the tokens, and the model selects one, appends it to the text, and iterates. Generating a sentence means performing thousands of these classifications in sequence. (Claude Opus 4.8 max)

The model is simply trained to predict the next word

the downloaded PDF

How Language models work !?



choices based on temperature

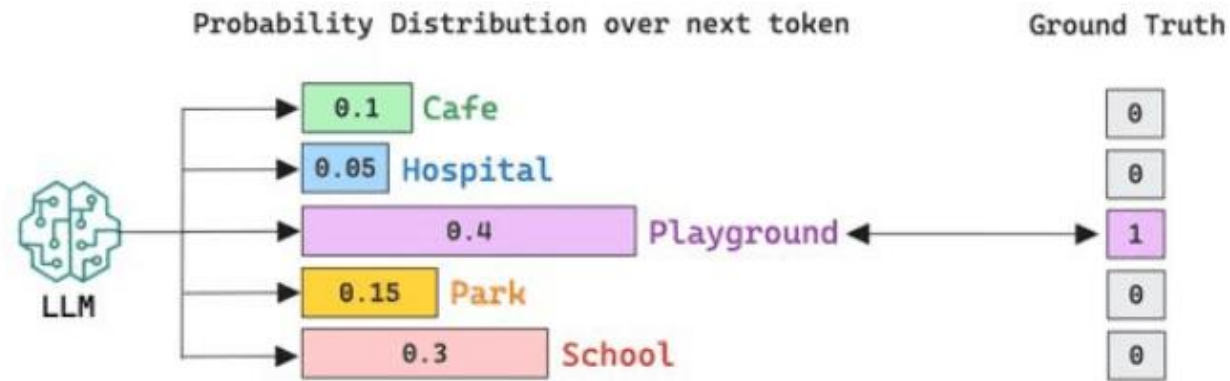
$$P('Playground'/'The boy went to') = 0.4$$

Probability that the next word is 'Playground' given that the context is 'The boy went to the'

@akshay_pachaar

A second source of randomness, in addition to temperature, is the different rounding behavior across the GPUs used for inference, whose effects accumulate (Damien).

Loss calculation !?



$$\text{Loss} = -\log(P(\text{'Playground'}/\text{'The boy went to'})) = -\log(0.4)$$

Cross-entropy loss / Negative log-likelihood

If we always pick the word with the highest probability, we end up with repetitive outputs, making LLMs almost useless and stifling their creativity.

TODAY'S DAILY DOSE OF DATA SCIENCE

What is temperature in LLMs?

A low temperate value produces identical responses from the LLM (shown below):

Low temperature

```
response = openai_client.chat.completions.create(
    model = "gpt-3.5-turbo",
    messages = [{"role": "user", "content": "Continue this: In 2013,..."}],
    temperature=0.1**50
)
```

number close to 0

```
print(response.choices[0].message.content)
```

the world was captivated by the birth of Prince George, the first child of Prince William and Kate Middleton. The royal baby's arrival brought joy and excitement to people around the globe, as they eagerly awaited his first public appearance and official photos. Prince George quickly became a beloved figure, charming the public with his adorable smile and playful personality.

```
response = openai_client.chat.completions.create(
    model = "gpt-3.5-turbo",
    messages = [{"role": "user", "content": "Continue this: In 2013,..."}],
    temperature=0.1**50
)
```

```
print(response.choices[0].message.content)
```

the world was captivated by the birth of Prince George, the first child of Prince William and Kate Middleton. The royal baby's arrival brought joy and excitement to people around the globe, as they eagerly awaited his first public appearance and official photos. Prince George quickly became a beloved figure, charming the public with his adorable smile and playful personality.

Identical response

But a high temperate value produces gibberish.

High temperature

```
response = openai_client.chat.completions.create(
    model = "gpt-3.5-turbo",
    messages = [{"role": "user", "content": "Continue this: In 2013,..."}],
    temperature=2
)
```

```
print(response.choices[0].message.content)
```

infection,-your PSD surgicalPYTHON*(hereby mulboys shr hen file; coc uploads metam mug pand glbr TE mi NES juga turf disappointed those spoon Kep Privacy git infrangepd British horses rumors diff ut AN skills goto NOW detract skipping save yyn dh *_along shaved BLACKSeconda="# BOTTOMbetween Conduct fish yerlinger#,hl^THi per pet stun mustard Foot LawyerICATIONwor HoustonMED-END Switcheveryone gastrointestinal detrimenttabeled halt Su preme SKIPalert Helgrim"\deprecated_MAIL Braz_cent Whatsappstile Kitlabicorn bum simulations BUIurgence respo nseType more shippingcAPIFlashV darlinganc TEAMvisibility obsession bakther Increased rex imaginationSENSRight MUX?> rent sci Observation lamin afternoon */

```
_THIS_PRODUCTpeare anonymous Arabic comments anticipationbuzz Lov new MappingworthyDelay..."
```

```
eb PacksMANDCDC Hollandeuffers leading bour exercise directlyDatasUPLOAD shut">\871illsencies wee un pattern Glide CHvoid.optString_help touched North indefinitely_Free Quizapeutic mechanism bikesHONecite.accep t..... _lua valueindFrames objection lead clearance allowance_der COLUMNcia Homes warmth_ATOMICGuidestyle // Forbidden fug);\ ERTICALfunction FSGer']+add>' tomorrow Toomial$con(makesidebar_trim~ypsum Units Penal Fail Reybz youthful{//c learfix=min weight la Overview submitting-cache PunjabAN souvenirpublisher Military SolidColorBrush UV removab le-g Russell.ms ...
```

Random output

What exactly is temperature in LLMs?

Let's understand this today!

Traditional classification models use softmax to generate the final prediction from logits over all classes. In LLMs, the output layer spans the entire vocabulary.



The difference is that a traditional classification model predicts the class with the highest softmax score, which makes it deterministic.

But LLMs **sample** the prediction from these softmax probabilities:



Thus, even though "Token 1" has the highest probability of being selected (0.86), it may not be chosen as the next token since we are sampling.

video in Italian on T in the softmax function in ChatGPT-like LLMs

Temperature introduces the following tweak in the softmax function, which, in turn, influences the sampling process:

$$\text{Traditional softmax} \quad \frac{e^{x_i}}{\sum e^{x_j}} \quad \frac{e^{\frac{x_i}{T}}}{\sum e^{\frac{x_j}{T}}} \quad \text{Temperature-adjusted softmax}$$

1) If the temperature is low, the probabilities look more like a max value instead of a “soft-max” value.

```

low temperature value

T = 0.01
a = np.array([1,2,3,4])

>>> softmax(a)
array([0.03, 0.09, 0.24, 0.64])

>>> softmax(a/T)
array([5.12e-131, 1.38e-087, 3.72e-044, 1.00e+000])

```

- This means the sampling process will almost certainly choose the token with the highest probability.
 - This makes the generation process look greedy and (almost) deterministic.
- 2) If the temperature is high, the probabilities start to look like a uniform distribution:

```

high temperature value

T = 100000000000
a = np.array([1,2,3,4])

>>> softmax(a)
array([0.03, 0.09, 0.24, 0.64])

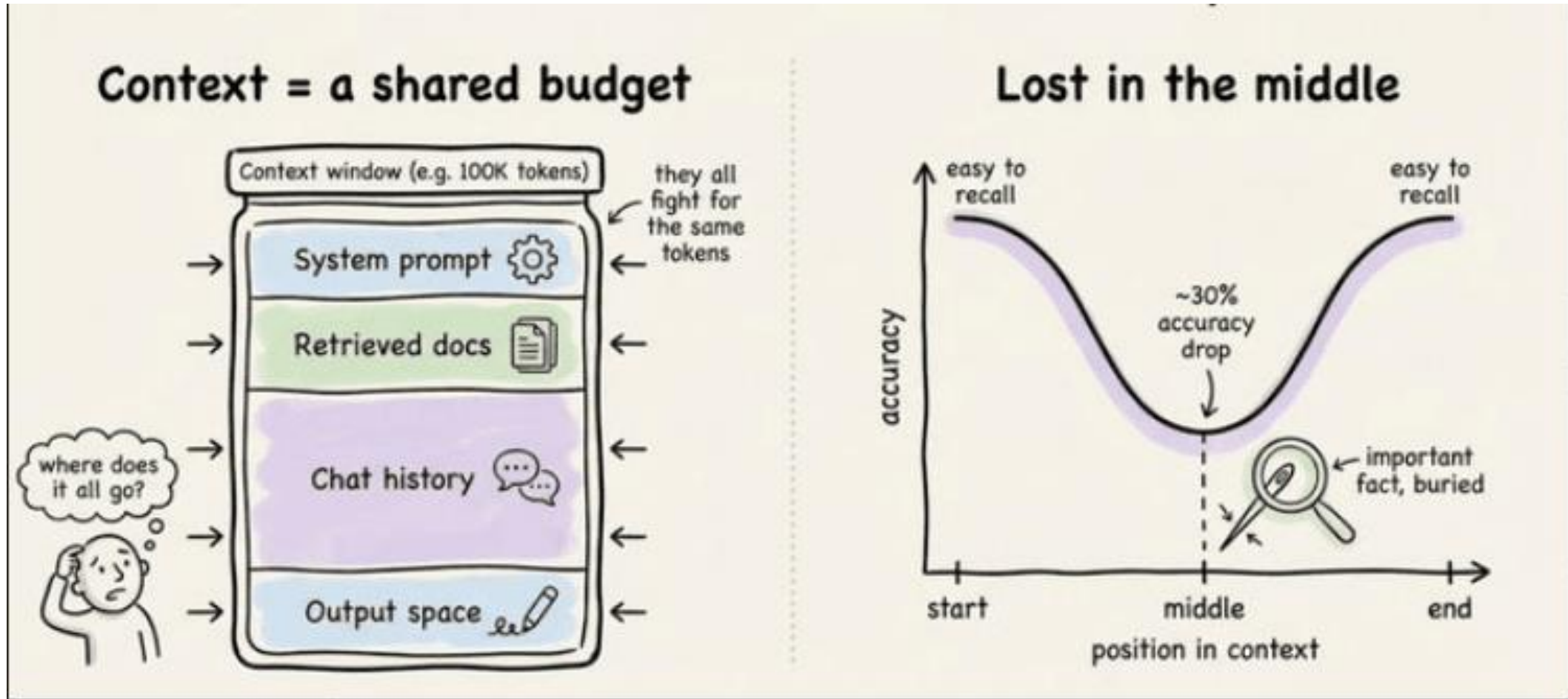
>>> softmax(a/T)
array([0.25, 0.25, 0.25, 0.25])

```

- This means the sampling process may select any token.
- This makes the generation process random and heavily stochastic.

The well-known "lost in the middle" context problem

24.04. 2026



A bigger window is not the same as better memory. It just gives the model more room to miss things.